PERCENTAGE

PERCENTAGE (प्रतिशत %)

PER + CENT
↓ ↓
divide by 100

$X\% = \frac{X}{100}$

प्रतिशत वह भिन्न है जिसका हर 100 हो।

PERCENT vs PERCENTAGE

PERCENT ⇓
जब % किसी known/unknow value के साथ आता है तो उसे PERCENT बुलाया जाता है।
e.g X% = X PERCENT
20% = 20 PERCENT

PERCENTAGE ⇓
जब % अकेला आता है तो उसे % से बुलाया जाता है।
% = PERCENtage

why we study % (आखिर क्यों) ?

e.g Their are three students A, B & C
by looking at table I it looks like C is better but wait and see Table II

I

	MARKS
A	45
B	80
C	90

TABLE II

	MARKS	mm	%
A	45	50	90%
B	80	100	80%
C	90	150	60%

is better

Note:- Table I was incomplete. Now in Table II we calculate marks of A, B, C out of 100 (%)

IInd Table देखने के बाद हमने सबके MARKS 100 में से निकाल लिए (%). अब हम Compare कर सकते हैं और बता सकते हैं कि A is better.

%

divide by 100 OR Compare with 100

PERCENTAGE CHANGE

$$\% \text{ CHANGE} = \frac{\text{WHAT WE COMPARE (जिसको)}}{\text{TO WHOM WE COMPARE (जिसके साथ)}} \times 100$$

Let take Two No $A = 120$, $B = 150$

(i) A is what % of B
or what % of B is A $= \frac{A}{B} \times 100 = \frac{120}{150} \times 100 = 80\%$
or what % is A of B → [जिसके साथ of आता है वो नीचे आता है।]

(ii) B is what % of A
or what % of A is B $= \frac{B}{A} \times 100 = \frac{150}{120} \times 100$
or what % is B of A
$= 125\%$ Ans.

(iii) By how much % A is less than B $= \frac{B-A}{B} \times 100$
$= \frac{30}{150} \times 100 = 20\%$

(iv) By how much % B is more than A
$= \frac{30}{120} \times 100 = 25\%$

Concept of By and To

1) A Number is decrease To 40% [उतना हो जायगा]

$\Rightarrow$ INITIAL 100 FINAL 40

2) A Number is decreased by (उतने से घट जायगा)

$\Rightarrow$ INITIAL 100 FINAL 60

PERCENTAGE CHANGE vs PERCENTAGE POINT CHANGE

$A = 10\%$ $B = 20\%$

% change $= \frac{5\%}{10\%} \times 100 = \underline{\underline{50\%}}$

$15\% - 10\% = \underline{\underline{5\%}}$

% To FRACTION [divide by 100] | FRACTION To % [multiply by 100]

$20\% = \frac{20}{100} = \frac{1}{5}$

$12.5\% = \frac{12.5}{100} = \frac{1}{8}$

$40\% = \frac{40}{100} = \frac{2}{5}$

$\frac{1}{5} \Rightarrow \frac{1}{5} \times 100 = 20\%$

$\frac{1}{6} \Rightarrow \frac{1}{6} \times 100 = 16\frac{2}{3}\%$

$\frac{3}{8} \Rightarrow \frac{3}{8} \times 100 = 37.5\%$

How To calculate % Quickly

520 → 100%

→ 10% = 52

→ 1% = 5.2

→ .1% = .52

21% of 520

$= 10\% + 10\% + 1\%$

$= 52 + 52 + 5.2$

$= 109.2$ ANS.

27% of 520

$= 25\% + 1\% + 1\%$

$= \frac{1}{4} \times 520 + 5.2 + 5.2$

$= 130 + 10.4 = 140.4$

57% of 520

$= 50\% + 5\% + 2\%$

$= 260 + 26 + 5.2 + 5.2$

$= 286 + 10.4$

$= 296.4$ ANS.

144 is what % of 450

$= \frac{144}{450} \times 100$

$= \frac{(135+9)}{450} \times 100$

$= \frac{135}{450} \times 100 + \frac{9}{450} \times 100$

$= 30\% + 2\% = 32\%$ Ans.

126 is what % of 450

$= \frac{126}{450} \times 100$

$= \frac{135-9}{450} \times 100$

$= \frac{135}{450} \times 100 - \frac{9}{450} \times 100$

$= 30\% - 2\% = 28\%$

138 is what % of 476

$= \frac{138}{476} \times 100$

$= \frac{(119+19)}{476} \times 100$

$= \frac{119}{476} \times 100 + \frac{19}{476} \times 100$

$= 25\% + 4\% \cong \underline{29\%}$

63% of 560

$= (62.5\% + .5\%)$ of 560

$= \frac{5}{8} \times 560 + 2.8$

$= 350 + 2.8$

$= 352.8$ ANS.

FRACTION TO PERCENTAGE CONVERSION TABLE

Fraction	Percentage I	Percentage II	Fraction	Percentage I	Percentage II
1	100%	100%	$\frac{1}{14}$	7.14%	$7\frac{1}{7}\%$
$\frac{1}{2}$	50%	50%	$\frac{1}{15}$	6.67%	$6\frac{2}{3}\%$
$\frac{1}{3}$	33.33%	$33\frac{1}{3}\%$	$\frac{1}{16}$	6.25%	$6\frac{1}{4}\%$
$\frac{1}{4}$	25%	25%	$\frac{1}{17}$	5.88%	$5\frac{15}{17}\%$
$\frac{1}{5}$	20%	20%	$\frac{1}{18}$	5.56%	$5\frac{5}{9}\%$
$\frac{1}{6}$	16.67%	$16\frac{2}{3}\%$	$\frac{1}{19}$	5.26%	$5\frac{5}{19}\%$
$\frac{1}{7}$	14.28%	$14\frac{2}{7}\%$	$\frac{1}{20}$	5%	5%
$\frac{1}{8}$	12.5%	$12\frac{1}{2}\%$	$\frac{1}{24}$	4.16%	$4\frac{1}{6}\%$
$\frac{1}{9}$	11.11%	$11\frac{1}{9}\%$	$\frac{1}{25}$	4%	4%
$\frac{1}{10}$	10%	10%	$\frac{3}{4}$	75%	75%
$\frac{1}{11}$	9.09%	$9\frac{1}{11}\%$	$\frac{2}{5}$	40%	40%
$\frac{1}{12}$	8.33%	$8\frac{1}{3}\%$			
$\frac{1}{13}$	7.69%	$7\frac{9}{13}\%$			

DERIVED FRACTION TO CONVERSION Table

Fraction	Percentage I	Percentage II	Fraction	Percentage I	Percentage II
$\frac{1}{8}$	12.5%	$12\frac{1}{2}$%	$\frac{2}{3}$	66.67%	$66\frac{2}{3}$%
$\frac{3}{8}$	37.5%	$37\frac{1}{2}$%	$\frac{1}{12}$	8.33%	$8\frac{1}{3}$%
$\frac{5}{8}$	62.5%	$62\frac{1}{2}$%	$\frac{7}{12}$	58.33%	$58\frac{1}{3}$%
$\frac{7}{8}$	87.5%	$87\frac{1}{2}$%	$\frac{11}{12}$	91.67%	$91\frac{2}{3}$%
$\frac{1}{6}$	16.67%	$16\frac{2}{3}$%	$\frac{5}{6}$	83.33%	$83\frac{1}{3}$%

Larger FRACTION/PERCENTages

1) 108.33% or $108\frac{1}{3}\% = 100\% + 8.33\% = 1 + \frac{1}{12} = \frac{13}{12}$

2) 362.5% or $362\frac{1}{2}\% = 300\% + 62.5\% = 3 + \frac{5}{8} = \frac{29}{8}$

3) 191.67% or $191\frac{2}{3}\% = 200\% - 8\frac{1}{3}\% = 2 - \frac{1}{12} = \frac{23}{12}$

4) 393.33% or $393\frac{1}{3}\% = 400\% - 6\frac{2}{3}\% = 4 - \frac{1}{15} = \frac{59}{15}$

5) 283.33% or $283\frac{1}{3}\% = 200\% + 83\frac{1}{3}\% = 2 + \frac{5}{6} = \frac{17}{6}$

OR $= 300 - 16\frac{2}{3}\% = 3 - \frac{1}{6} = \frac{17}{6}$

PERCENTAGE IS INTERCHANGABLE

$A\%$ of B = $B\%$ of A = $A \times B/100$

e.g. 64% of 62.5 = ?

= 62.5% of 64 [∵ % is interchangable]

= $\frac{5}{8} \times 64 = 40$ [∵ $62.5\% = \frac{5}{8}$]

e.g. 72% of $91\frac{2}{3}$ = ?

= $91\frac{2}{3}\%$ of 72

= $\frac{11}{12} \times 72 = 66$ ANS. [∵ $91\frac{2}{3}\% = \frac{11}{12}$]

$17\frac{1}{2}\%$ of 84 = ?

= 35% of 42 [∵ $a\%$ of b = $2a\%$ of $\frac{b}{2}$]

= 70% of 21

= 14.7 ANS.

32% of $250 + 12.5\%$ of 640 = ?

12.5% of 640 = 25% of 320 [∵ $a\%$ of b = $2a\%$ of $\frac{b}{2}$]

∴ 32% of $250 + 25\%$ of 320

= 25% of $320 + 25\%$ of 320

= $(25\% + 25\%)$ of 320 = 50% of 320 = 160 ANS.

62.5% of $512 + 83\frac{1}{3}\%$ of $216 = ?$

$= \frac{5}{8} \times 512 + \frac{5}{6} \times 216$ $\left[\because 62\frac{1}{2}\% = \frac{5}{8}\right.$,

$\left. 83\frac{1}{3}\% = \frac{5}{6}\right]$

$= 5 \times 64 + 5 \times 36 = 5(100)$

$= 500$ ANS.

$193\frac{1}{3}\%$ of $225 + 91\frac{2}{3}\%$ of $144 = ?$

$\left(200\% - 6\frac{2}{3}\%\right) 225 + \left(100\% - 8\frac{1}{3}\%\right)$ of 144

$= 450 - \frac{1}{15} \times 225 + 144 - \frac{1}{12} \times 144$

$= 435 + 132 = 567$ $\left[\because 6\frac{2}{3}\% = \frac{1}{15},\right.$

$\left. 8\frac{1}{3}\% = \frac{1}{12}\right]$

Population of city beautiful chandigarh is 490000 in 2016. If growth Rate is $14\frac{2}{7}\%$. What will be Population in 2017?

Solution :— $P_{2016} = 490000$

$14\frac{2}{7}\%$ of $490000 = \frac{1}{7} \times 490000 = 70000$

$P_{2017} = 490000 + 70000 = 560000$

Tricky Concepts

1. Convert % into fraction $\pm\frac{N}{D}$

   where + means increase

   & − means decrease.

2. Initial value = Denominator = D

3. Final value = $D \pm N$ or $I \pm N$

   + for Increase

   − for Decrease

e.g. 20% Increase $= +\frac{1}{5}$ $1 \rightarrow$ Increase, $5 \rightarrow$ Initial

Final = 5+1 = 6

20% decrease $= -\frac{1}{5}$ $1 \rightarrow$ decrease, $5 \rightarrow$ Initial

Final = 5−1 = 4

If $16\frac{1}{3}\%$ of a Number is added with itself, resultant number becomes 3430. Find the original Number?

<u>Solution</u> :− $16\frac{2}{3}\% \uparrow = +\frac{1}{6}$ $6 \rightarrow I$

$\Rightarrow F = 6 + 1 = 7$

$7 \longrightarrow 3430$

$\Rightarrow 6 \longrightarrow \frac{3430}{7} \times 6 =$ 2940 <u>Ans.</u>

If $16\frac{2}{3}\%$ of a Number is substracted from itself then the resultant Number becomes 225. Find the original Number?

Sol. $16\frac{2}{3}\% = \frac{-1 \to \text{Dec.}}{6 \to \text{INITIAL}} \Rightarrow \text{Final} = 6-1=5$

$5 \to 225$

$1 \to 45 \Rightarrow 6 \to 45\times 6 = 270$ **ANS.**

IF 210 is added in a no. then the no. becomes 137.5% of itself. Find the original Number?

Solution :- $137\frac{1}{2}\% = 100\% + 37\frac{1}{2}\% = 1+\frac{3}{8} = \frac{11 \to F}{8 \to I}$

Final $\leftarrow 11$, Initial $\leftarrow 8$; $3 \to 210$, $1 \to 70$ $\longrightarrow$ original $= 8 \longrightarrow 8\times 70$

560 **ANS.**

Population of a city is 72900 in 2010. If population growth rate is $12\frac{1}{2}\%$. Find population in 2009.

Solution :- $12\frac{1}{2}\% = \frac{+1 \to \text{Increase}}{8 \to 2009} \Rightarrow P_{2010} = 8+1=9$

$9 \to 72900$

$1 \to 8100$ $\quad P_{2009} = 8 \longrightarrow 8\times 8100 = 64800$

SUCCESSIVE (क्रमागत / लगातार) % Change

$\text{I} \xrightarrow{x\%} \bigcirc \xrightarrow{y\%} \text{F}$ $\qquad$ $\text{I} \xrightarrow{x\%} \bigcirc \xrightarrow{y\%} \bigcirc \xrightarrow{z\%} \text{F}$

$\% \text{ change} = x + y + \frac{xy}{100}$ $\qquad$ $= x + y + z + \frac{xy + yz + zx}{100} + \frac{xyz}{100^2}$

IF ↑ take x, y +ve

↓ take x, y −ve

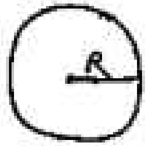
L ↑ = 20% B ↓ 10%

A = ?

$= 20 - 10 - \frac{200}{100}$

$= 8\% \uparrow$

Circle (वृत्त)



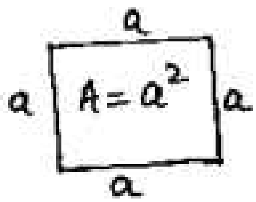
$A = \pi R^2$

R ↓ 10%

$A = -10 - 10 + \frac{100}{100}$

$= -19\% \downarrow$

Square

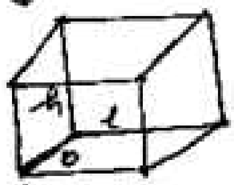


a ↑ 20%

$A = 20 + 20 + \frac{400}{100}$

$= 44\% \uparrow$

Cuboid



$V = lbh$

l ↑ 10% b ↓ 10%

h ↑ 20%

$V = 10 - 10 + 20 + \frac{-100 - 200 + 200}{100} + \frac{-10 \times 10 \times 20}{100^2}$

$= 20 - 1 - \frac{1}{5} = \frac{94}{5} = 18.8\% \uparrow$

50% + 40% = ?

$= -50 - 40 + \frac{2000}{100}$

$= -70\%$ ↓

Shows discount

30% + 20% + 10% = ?

$= -30 - 20 - 10 + \frac{600 + 200 + 300}{100} - \frac{6000}{100^2}$

$= -60 + 11 - .6$

$= -49.6$

Ans 49.6%

If the length of reactangle is increased by $12\frac{1}{2}\%$ and breadth decreased by 20%. Find the % change in Area?

Solution :-

	L	× B	= A
original	8	5	= 40
New	9	4	= 36

$12\frac{1}{2}\% = +\frac{1}{8}$

$20\% \downarrow = -\frac{1}{5}$

$\% \text{ change} = -\frac{4}{40} \times 100 = -10\%$ Ans

− shows Area is ↓

If radius of circle is decreased by 10%. Find the % change in Area of circle. $(A = \pi R^2)$

Solution :-

	Radius	Area
original	10	100
New	9	81

$\left.\begin{matrix}100\\81\end{matrix}\right\} -19$

$10\% \downarrow = -\frac{1}{10}$

$\text{New} = 10 - 1 = 9$

$\% \text{ change} = -\frac{19}{100} \times 100 = 19\% \downarrow$

If the price of the sugar is increased by $16\frac{2}{3}\%$ and consumption is decreased by 25%. find the % change in Expenditure?

Solution

	Price ×	Consump.	= Exp.
original	6	4	= 24
New	7	3	= 21

$\left.\begin{matrix}24\\21\end{matrix}\right\} -3$

$16\frac{2}{3}\% \uparrow = +\frac{1}{6}$

$25\% \downarrow = -\frac{1}{4}$

$\% \text{ change} = \frac{-3}{24} \times 100 = 12.5\% \downarrow$ Ans.

The value of a machine depreciates at the Rate of 10% per annum. If its present worth is 364000. find its worth after 3 years

Solution $10\% \downarrow = -\frac{1}{10}$

$$\begin{array}{ccc} 10 & — & 9 \\ 10 & — & 9 \\ 10 & — & 9 \\ \hline 1000 & — & 729 \end{array}$$

$1000 \rightarrow 364000 \Rightarrow 729 \rightarrow 265356$ ANS.

The present population of a town is 10800. During the 1st yr. population increases by $66\frac{2}{3}\%$, while decreases by $16\frac{2}{3}\%$ during second year. During 3rd year increases by $33\frac{1}{3}\%$. Find the population of town 3 yrs hence?

Sol $66\frac{2}{3}\% \uparrow = +\frac{2}{3}$, $16\frac{2}{3}\% \downarrow = -\frac{1}{6}$, $33\frac{1}{3}\% = +\frac{1}{3}$

$$\begin{array}{cc} 3 & 5 \\ 6 & 5 \\ 3 & 4 \\ \hline 54. & 100 \end{array}$$

$54 \longrightarrow 10800$

$\Rightarrow 100 \longrightarrow \boxed{20,000}$ ANS.

OR 3

6

3

The single discount which is equivalent to successive discounts of 50% & 40% ?

Solution :-

	50%	40%		
original	2	5	= 10	7
New	1	3	= 3	

equivalent d% = $-7/10 \times 100$ = 70% Ans.

discount is always decrease

$50\% \downarrow = -\frac{1}{2}$

$40\% \downarrow = -\frac{2}{5}$

A single discount which is equivalent to successive discounts of 30%, 20% & 10% ?

Solution

	30%	20%	10%		
original	10	5	10	= 500	– 248
New	7	4	9	= 252	

equivalent discount % = $\frac{248}{500} \times 100 = 49.6\%$

If the price of cinema ticket is increased by $16\frac{1}{3}\%$, then the sale will be decreased by 20%. Find the % change in Revenue ?

Solution :-

	Price	× Sale	= Revenue	
original	6	5	= 30	– 2
New	7	4	= 28	

$16\frac{1}{3}\% \uparrow = +\frac{1}{6}$

$20\% \downarrow = -\frac{1}{5}$

% change $= \frac{-2}{30} \times 100 = 6\frac{1}{2}\% \downarrow$

Ans.

Product Constancy method

Let $P = A \times B$

If one increases by some %, in order to keep P as constant, he should decrease B or vice-versa.

or in simple way If $P =$ constant

$\Rightarrow A \propto \frac{1}{B}$ [A & B are inversely proportional]

e.g. If $A = 2:3$, & If $P =$ constant

$\Rightarrow B = 3:2$

Application

(i) Area of Reactangle $A = l \times b$

(ii) Expenditure = Price x Consumption

$E = P \times C$

(iii) Revenue = Price x Sales, $R = P \times S$

(iv) Distance = Speed x Time, $D = S \times T$

(v) WORK = Time x Efficiency etc

If the price of sugar is increased by 20%, then by how much % the consumption should be ↓ decreased so that expenditure will remain same?

Solution :- $20\% \uparrow = +\frac{1}{5}$ → Increase, 5 → old Price

New Price $= 5+1 = 6$

Also $E = P \times C$ as $E =$ constant $\Rightarrow P \propto \frac{1}{C}$

	original		New
Price →	5	:	6
⇒ Consumption →	6	:	5

$[\because P \propto \frac{1}{C}]$

$\underbrace{6 : 5}_{-1}$

$\% \downarrow = -\frac{1}{6} = 16\frac{2}{3}\%$ ANS.

If the length of reactangle is decreased by 12.5%. By how much percentage breadth must be decreased in order to keep Area same?

Solution :- $12.5\% \downarrow = -\frac{1}{8}$ → old length

New length $= 8-1 = 7$

$A = l \times b$, as $A =$ Constant $\Rightarrow l \propto \frac{1}{B}$

	old		New
length	8	:	7
Breadth	7	:	8

$[\because l \propto \frac{1}{B}]$

$\underbrace{7 : 8}_{+1}$

$\% \uparrow = +\frac{1}{7} = 14\frac{2}{7}\%$ ANS.

\# A reduction ~~in~~ of 25% in the Price of sugar enables a housewife to purchase 4 kg more for ₹ 800. Find original and current Price per kg?

Solution :→ Here Expenditure = Constant = 800

$\Rightarrow$ Price $\propto \frac{1}{\text{Consumption}}$

$25\%\downarrow = -\frac{1}{4} \rightarrow$ old $\Rightarrow$ New $= 4-1 = 3$

	old		New
Price	4	:	3
Consump	3	:	4

$\underbrace{3 : 4}_{+1 \rightarrow 4}$ $\Rightarrow$ old consumption $= 3\times4 = 12$ kg

New Consumption $= 4\times4 = 16$ kg

OR DIRECT
25% of 800 = 200
New/Reduced Price $= \frac{200}{4} = 50$
old $\Rightarrow$ 75% → 50
100% → $66\frac{2}{3}$

(i) old Price $= \frac{800}{12} = 66\frac{2}{3}$ ₹/kg (ii) New/current Price $= \frac{800}{16} = 50$

\# Due to 30% increase in Price of apples, 6 apples are less available for ₹ 520. Find the old and new Price of a apple?

Solution :– $30\%\uparrow = +\frac{3}{10}$

	old		New
Price	10	:	13
Cons.	13	:	10

$\underbrace{13 : 10}_{-3 \rightarrow 6}$

$1 \rightarrow 2$

OR DIRECT
30% of 520 = 156
New Price $= \frac{156}{6} = 26$
old → 130% — 26
100% → 20

old Cons. = 13×2 = 26

old Price $= \frac{520}{26} = 20$₹

New Consumpt. = 10×2 = 20

New Price $= \frac{520}{20} = $ ₹ 26

A reduction of 20% in the price of sugar enables a man to buy 10 kg more for ₹ 54. Find the original as well as reduced price per kg?

Solution :→ Here Expenditure = 54 = constant

$\Rightarrow$ price $\propto \frac{1}{\text{Consumption}}$

$20\% \downarrow = -\frac{1}{5}$

	old		New
price =	5	:	4
Consumption =	4	:	5

$+1 \longrightarrow 10\text{ kg}$

$\Rightarrow$ old Consum = $10 \times 4 = 40$

New " = $10 \times 5 = 50$

Old Price = $\frac{54}{40} = 1.35$ ₹/kg

New or Reduced = $\frac{54}{50} = 1.08$ ₹/kg

OR DIRECT

20% of 54 = 10.8

New/Reduced Price = $\frac{10.8}{10}$

= 1.08 Ans.

Due to an increase ~~in~~ of 20% in the price of eggs, 2 eggs less available for ₹ 24. The present rate of egg per dozen?

Solution :→ E = constant

$20\% \uparrow = +\frac{1}{5}$

	old		New
Price	5	:	6
Consump.	6	:	5

$-1 \longrightarrow 2$

New Consumption

$\Rightarrow 5 \rightarrow 5 \times 2 = 10$ egg

Price per egg = $\frac{24}{10} = 2.4$

Price of dozen = 2.4×12 = 28.80 ₹ Ans.

If the Price of sugar is decreased by 40%. By how much % consumption should be increased so that expenditure will decrease by 10%?

Solution :→ METHOD→1 $E = P \times C$

Exp. = Price × Consumption

Let old $100 = 10 \times 10$ (-10%, -40%)

New $90 = 6 \times$ New Consumption

$\Rightarrow$ New Consumption $= \frac{90}{6} = 15$

% ↑ in Consumption $= \frac{15-10}{10} \times 100 = 50\%$ Ans.

OR

$= x + y + \frac{xy}{100}$

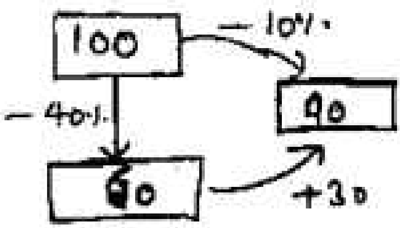
$-10 = -40 + x - \frac{40x}{100}$

$\Rightarrow 60x = 3000$

$\Rightarrow x = 50\%$

50% ↑

METHOD 2 :—



$\Rightarrow$ % Change $= \frac{30}{60} \times 100$

$= 50\%$

If the Price of Rice is decreased by 20%. By how much % consumption should be increased so that Expenditure also Increased by 20%?

Solution :— 1st Method

$E = P \times C$

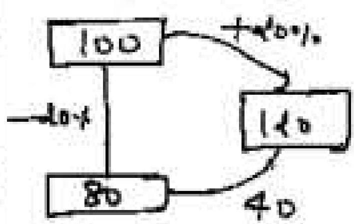
Let old $100 = 10 \times 10$

New $120 = 8 \times N_C$

$\Rightarrow N_C = 15$

% ↑ $= \frac{5}{10} \times 100 = 50\%$

2nd Method



% increase

$= \frac{40}{80} \times 100 = 50\%$

Focus on Fix :→ The quantity that does not change in initial and final mixture make that equal.

e.g. If there is mixture of water and salt & water get evaporated. That means salt will not change in initial and final mixture. make salt equal. You will better understand by questions.

A vessel has 60 litres of solution of acid & water having 80% acid. How much water must be added to make it a solution in which acid is 60%?

Solution :- water is added ⇒ acid qty is same

1st METHOD

let final mixture (after add water) is = x ltr

equate Acid

$$\frac{80 \times 60}{100} = \frac{60 \times x}{100}$$

$\Rightarrow x = 80$ ltr

water added = 80 − 60 = 20

2ND METHOD

by Ratio

	Acid		water
old	$4_{\times 3}$	:	$1_{\times 3}$
New	$3_{\times 4}$	:	$2_{\times 4}$

make Acid equal

12 : 3
12 : 8

5 water added

$15 \rightarrow 60 \Rightarrow 5 \rightarrow \boxed{20 \text{ ltr}}$

75 gm of a solution has 30% sugar in it. Then the qty of sugar that should be added to the solution to make 70% sugar solution. $\boxed{\text{Ans. 100 gm}}$

FRESH grapes contains 90% water while dry grapes contains 20% water. what is the weight of dry grapes obtained from 20 kg FRESH GRAPES.?

Solution 1st method

the weight of non-water part called pulp will Remain same equate.

$$\frac{20*10}{100} = \frac{80*X}{100} \Rightarrow X = 2.5 \text{ kg}$$

$x \rightarrow$ dry grapes weight.

2nd METHOD Ratio

	PULP		WATER
old	$1_{\times 4}$	:	$9_{\times 4}$
New	4	:	1

make PULP same

~~5 : 45~~ $4 : 36 \rightarrow 40$

~~5 :~~ $4 : 1 \rightarrow 5$

$40 \rightarrow 20$ kg $\Rightarrow 5 \rightarrow \boxed{2.5 \text{ kg}}$

FRESH GRAPES contains 80% water, while dry grapes contains 10% water. If the weight of dry grapes is 500 kg. what is total wt. when it is FRESH?

Solution :- equate Non water Part (Pulp)

1st method

let total wt. of fresh grapes $= x$ kg.

Pulp$_1$ = Pulp$_2$

$$\frac{20*X}{100} = \frac{90*500}{100}$$

$\Rightarrow x = 2250$ Kg.

2nd METHOD (Ratio)

	PULP		WATER
old	$1_{\times 9}$	:	$4_{\times 9}$
New	9	:	1

Dry $10(9+1) = 500$

$1 \rightarrow 50$ kg

FRESH $= 9 + 36 = 45$

wt $= 45 \times 50 = 2250$ kg

ELECTIONS RELATED QUESTIONS

- Total votes (कुल मत) in voter list
  - votes casted/polled (डाले गए मत)
    - Valid Votes (वैध मत)
      - winner (जीतने वाला)
      - LOSER (हारने वाला)
    - Invalid Votes (अवैध मत)
  - Votes not polled (ना डाले गए मत)

Note:- जो counting होगी वो सिर्फ valid votes की होगी OR we can say that difference of winners and losers votes is of valid votes only (winner + loser votes = valid votes)

Note: अगर 1 के वोट Total votes की % में given है तो Total votes को 100 मान लो और अगर valid votes की % में given है तो valid votes को 100 मानने में फायदा होगा।

\# In an election between two candidates, the candidate getting 60% of votes polled is elected by a majority of 14000 votes. The No. of votes get by winner 1) 28000 2) 32000 3) 42000 4) 46,000

Solution :- winner LOSER

60% $\quad$ 100 − 60 = 40%

$60\% - 40\% = 14000 \Rightarrow 20\% \rightarrow 14000$

$\therefore 60\% \rightarrow \frac{14000}{20} \times 60$

$= 42000$ ANS

\# एक चुनाव में दो उम्मीदवार चुनाव लड़ रहे थे। जीतने वाले उम्मीदवार ने कुल मतों का 55% प्राप्त हुआ और वह 500 मतों से चुनाव जीत गया, तो उस चुनाव में कुल कितने वोट डाले गए।

Solution :- W $\quad$ L

55% $\quad$ 45%

(margin) $10\% \rightarrow 500$

$\Rightarrow 100\% \rightarrow 5000$ ANS.

\# In an election b/w BJP & CONGRESS, BJP SCORED 22% of total votes more than Cong. if congress get 11700 votes and their is no invalid vote then find the total No. of votes. [30000 ANS.]

8% of the voters in an election did not cast their votes. In this election their were only two candidates. The winner by obtaining 48% of total votes defeated his opponent by 1100 votes. The total no. of voters in voter list is a). 21000 b). 23500 c). 24000 d). 27500

Solution :- Let Total votes = 100

Total votes $\boxed{100}$ → कुल मत

(invalid) −8 (अवैध)

(valid vote) $\boxed{92}$ (वैध मत)

W ↙ ↘ L

48 44

4 — 1100 ⇒ 100 → $\frac{1100}{4} \times 100 = 27500$ ANS.

[LOSER + WINNER = Valid vote

LOSER + 48 = 92

⇒ L = 92 − 48 = 44]

एक चुनाव में दो उम्मीदवार है। जिसमें 20% मतों को अवैध घोषित कर दिया और विजेता को कुल मतों का 56% वोट प्राप्त होते है तथा वह 6400 मतों से चुनाव जीत जाता है। तो चुनाव में कुल कितने मत/वोट डाले गए?

Solution W = 56% of 100 = 56

L + W = 80 ⇒ L = 24

$\boxed{100}$ → कुल मत

−20 (अवैध)

$\boxed{80}$ वैध

W ↙ ↘ L

56 24

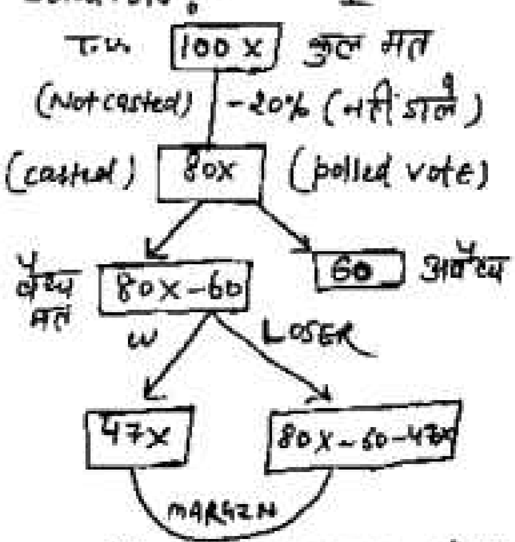
6400 ← 32

100% → $\frac{6400}{32} \times 100 = 20{,}000$

In an election their are two candidates 20% of voters didn't cast their votes. And 60 votes were declared Invalid. If winners gets 47% of total votes and win by 200 votes then find the total no. of votes?

Solution :-

I

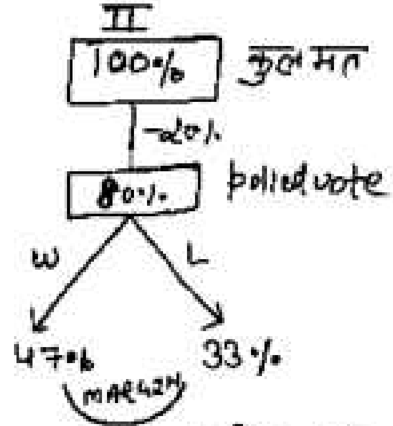


$47x - 80x + 60 + 47x = 200$

$\Rightarrow 14x = 200 - 60 = 140$

$\Rightarrow x = 10 \quad T.V = 100x$
$= 1000$ ANS.

II



14% = 200 − 60
100% = 1000 ANS

Note :- आपको invalid मतों को यहाँ - करना है ना कि ऊपर।

एक चुनाव में दो उम्मीदवार हैं। 20% मतदाताओं ने अपने मत का प्रयोग नहीं किया। और 200 मत अवैध पाये गये। जीतने वाले प्रत्याशी को कुल मतों का 48% प्राप्त हुआ और वह 6600 मतों से चुनाव जीत गया तो मतदाता सूची में कितने मतदाता थे?

[40000 ANS]

In an election two candidate participated. 10% votes declared invalid and the winner gets 60% of the valid votes and win by 5400 votes find the total No. of votes in voting list?

Solution :- I

voting list $\boxed{100}$ कुल मत

| -10% invalid

(valid vote) $\boxed{90}$ वैध मत

W 60% / 40%

54   36

MARGIN

$18 \longrightarrow 5400$

$100 \longrightarrow \frac{5400}{18} \times 100$

$= 30{,}000$ Ans.

II

Let Total votes x

$\therefore x * \frac{9}{10} =$ valid votes

$W = 60\%$ of valid votes

$\Rightarrow L = 40\%$ of valid votes

margin $= 20\% = \frac{1}{5}$ of valid votes

$\therefore \boxed{x * \frac{9}{10} * \frac{1}{5} = 5400}$

$\Rightarrow x = 30{,}000$ Ans.

एक चुनाव में दो प्रत्याशी चुनाव लड़ रहे हैं। उसमें 20% मतों को अवैध घोषित किया गया। विजेता को कुल वैध मतों का 60% मत प्राप्त हुए और वह 4096 मतों से चुनाव जीत गया तो उस चुनाव में कुल कितने मत डाले गए।

Solution :- I

$\boxed{100}$ कुल मत

| -20% अवैध

$\boxed{80}$ वैध मत

60% W   L 40%

48   32

MARGIN

$16 — 4096$

$\Rightarrow 100 — \frac{4096}{16} \times 100$

$= 6400$ Ans.

$= 25600$

II

Let total votes $= (x)$

$\therefore x * \frac{80}{100} = x * \frac{4}{5} =$ वैध मत

विजेता 60% of वैध मत

$\Rightarrow$ LOSER $= 40\%$ of वैध मत

margin $= 20\%$ of वैध मत

$\therefore x * \frac{4}{5} * \frac{1}{5} = 4096$

$x = 25600$ Ans.

In an election two candidate participated. 10% voters did not vote, out of which 10% votes declared invalid and the winner got 60% of valid votes and win by 8100 votes. Then find the total no of votes in voter list?

Solution :- I

कुल मत $\boxed{100}$

-10%

Polled $\boxed{90}$

90% V → वैध $\boxed{81}$; 10% I → $\boxed{9}$

W 60% , L 40%

margin 20%

$\therefore 81 \times \frac{20}{100} = 8100$

$\Rightarrow 100 \rightarrow 50000$ ANS.

II

Let valid vote = 100

वैध $\boxed{100}$

W 60 , L 40

margin

$20 \rightarrow 8100$

$\Rightarrow 100 \rightarrow \frac{8100}{20} \times 100$

वैध मत = 40500

$T.V. = 40500 \times \frac{10}{9} \times \frac{10}{9}$

$= 50,000$ ANS.

III

Let T.V. = X

vote polled $= X * \frac{9}{10}$

valid votes $= X * \frac{9}{10} * \frac{9}{10}$

margin $= 20\% = \frac{1}{5}$

$\therefore X * \frac{9}{10} * \frac{9}{10} * \frac{1}{5}$

$= 8100$

$X = 50,000$ ANS.

एक चुनाव में दो प्रत्याशी थे। जिसमें 20% मतदाताओं ने अपने मत का प्रयोग नहीं किया। और 10% मत अवैध पाए गए। अगर विजेता को वैध मतों का 70% मिले तो वह 2160 मतों से चुनाव जीत जाता है। तो मतदाता सूची में कुल कितने मतदाता थे?

Solution : Let कुल मत = X

poll हुए वोट $= X * \frac{80}{100} = X * \frac{4}{5}$

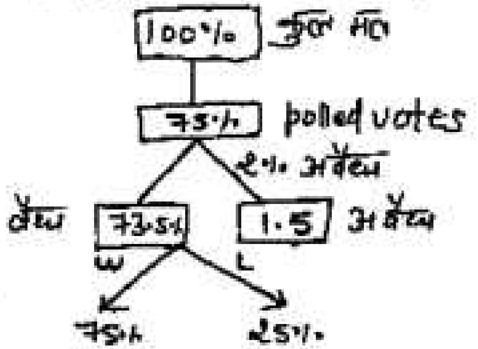
वैध मत $= X * \frac{4}{5} * \frac{9}{10}$

margin $= 70\% - 30\% = 40\%$ of वैध मत

$\therefore X * \frac{4}{5} \times \frac{9}{10} \times \frac{2}{5} = 2160 \Rightarrow X = 7500$ ANS.

In an election between two candidates 75% of the voters casts their vote, out of which 2% votes declared invalid. A candidate got 9261 votes which were 75% of valid votes. The total No. of voters in voting list was 1) 16000 2) 16400 c) 16800 4) 18000

Solution :- I



$\therefore$ 75% of 73.5% = 9261

$\Rightarrow$ 100% = 16800 Ans.

II

माना कुल मत = X

Polled votes $= X * \frac{75}{100} = X * \frac{3}{4}$

वैध मत (valid) $= X * \frac{3}{4} * \frac{98}{100}$

winner get 75% of valid vote

$\therefore \left[X * \frac{3}{4} * \frac{98}{100}\right] * \frac{75}{100} = 9261$

$\Rightarrow X = 16800$ Ans.

एक चुनाव में बंसीलाल और देवीलाल ने भाग लिया। 37.5% मतदाताओं ने बंसीलाल को वोट करने का वादा किया और शेष ने देवीलाल को वादा किया। वोट वाले दिन 20% बंसीलाल के वोटरों ने और 25% देवीलाल के वोटरों ने अपना वादा तोड़ दिया (मुकर गए)। कुल कितने वोट थे अगर देवीलाल 8400 वोटों से चुनाव जीत गया?

Solution :-

$37.5\% = \frac{3}{8}$

माना Total वोट 800

$\Rightarrow$ देवीलाल $\rightarrow$ 500

बंसीलाल $\rightarrow$ 300

बंसीलाल	देवीलाल
300	500
−60 (20%)	−125
240	375
+125	+60
365	435

70 — 8400

$\rightarrow$ 800 $\rightarrow$ 96000 Ans

किसी चुनाव में दो उम्मीदवार हैं। 10% मतदाताओं ने वोट नहीं डाले और 500 वोट अवैध घोषित कर दिए जाते हैं। जीतने वाला उम्मीदवार वैध मतों का 60% प्राप्त करता है और 800 वोटों से चुनाव जीत जाता है। पंजीकृत मतदाताओं की संख्या ज्ञात करें।

Solution :- I

Let valid vote = 100

$\boxed{100}$ वैध मत

W → 60, L → 40

MARGIN

$20 \longrightarrow 800$

$100 \longrightarrow \frac{800}{20} \times 100$

वैध मत = 4000

votes Polled = वैध + अवैध

= 4000 + 500 = 4500

Let x = T.V. = कुल मत

$\therefore \; x * \frac{90}{100} = 4500 \Rightarrow x = 5000$ Ans.

II

Let कुल मत = x

वोट पड़े = $x * \frac{90}{100} = x * \frac{9}{10}$

वैध मत = $x * \frac{9}{100} - 500$

margin = 60% − 40% = 20%

= $\frac{1}{5}$ of वैध मत

$\therefore \left[x * \frac{9}{100} - 500\right] * \frac{1}{5} = 800$

$\Rightarrow x = 5000$ Ans.

In an election two candidate participated. 20% voters did not cast their votes out of which 1600 votes declared invalid and the winner get 62.5% of valid votes and wins by 2000 votes. Find the number of votes in voting list?

Solution :- Let Total votes = x

$\therefore \left[x * \frac{4}{5} - 1600\right] \times \frac{1}{4} = 2000$

$\Rightarrow x = 12000$ Ans.

W 62.5% L 37.5%

25% = $\frac{1}{4}$

MARGIN

एक आदमी अपनी आय का 75% खर्च करता है। यदि उसकी आय में 20% की और खर्च में 10% की वृद्धि हो जाए तो उसकी बचत में कितने प्रतिशत की वृद्धि होगी?

Solution :- I (BASIC)

INCOME − EXP = Saving

$100 - 75 = 25$

(+20% ↑) (10% ↑) x%

$120 - 82.5 = 37.5$

$x\% = \frac{12.5}{25} \times 100 = 50\%$

II (Calligation)

खर्च : बचत

75% : 25% = 3 : 1

E 10% S x%

I (20%)

3 : 1

$\Rightarrow \frac{x-20}{20-10} = \frac{3}{1}$

$\Rightarrow x = 50\%$

एक व्यक्ति अपनी आय का 20% किराए पर, शेष का 30% खाने पर और उसके शेष का 50% शिक्षा पर खर्च करता है। अन्त में उसकी बचत 6300/- है। तो उसकी आय क्या थी?

Solution :-

$$6300 \times \frac{100}{80} \times \frac{100}{70} \times \frac{100}{50} = 22500 \text{ ANS.}$$

कोई विक्रेता कुल बिक्री पर 12% कमीशन देता है, तथा 15000 रु से ऊपर की बिक्री पर 1% बोनस देता है। यदि विक्रेता की कुल आय 7650/- रु है तो कुल कितने रु की बिक्री की?

Solution :- I (BASIC)

माना कुल बिक्री = x रु

$\therefore x \times \frac{12}{100} + (x - 15000)\frac{1}{100} = 7650$

$\Rightarrow \frac{12x}{100} + \frac{x}{100} - 150 = 7650$

$\Rightarrow x = 60,000$ ANS.

II (TRICKY)

15000	+
12%	13%

1%

$15000 \times 1\% = 150$

$7650 + 150 = 7800$

$\Rightarrow 13\% \xrightarrow{\times 600} 7800$

$100\% \xrightarrow{\times 600} 60,000$

A student multiplied a Number by $\frac{3}{5}$ instead of $\frac{5}{3}$. What is the percentage error in the calculation?

Solution : Let the Number = 15 (LCM of 3 & 5)

wrong Result $= 15 \times \frac{3}{5} = 9$, Actual Result $= 25$

$\%\ \text{ERROR} = \frac{25-9}{25} \times 100 = 64\%$ Ans.

Two Number are less than a third Number by 30% and 37% respectively. The Percent by which second Number is less by first is?

Solution :– Let third No = 100

$\Rightarrow$ I No = 70 , II No = 63

Req. Result $= \frac{7}{70} \times 100 = 10\%$ Ans.

If the Numerator of a fraction is ↑ by 20% and denominator is ↓ by 10%. the resultant fraction becomes $\frac{36}{45}$. Find the original fraction?

Solution :– Let original fraction $= \frac{x}{y}$

$\therefore \quad \frac{x\,[1.20]}{y\,[.90]} = \frac{36}{45} \Rightarrow \frac{x}{y} = \frac{36}{45} \times \frac{9}{12} = \frac{3}{5}$ Ans.

If the Income Tax is increased by 19%, then Net income will be decreased by 6%. Find the Rate of income tax?

Solution :- [CONCEPT] Net Income = Income − Tax

$\Rightarrow$ Net Income + Tax = INCOME

in these type of questions Income is same

Note :- If Tax is ↑ by ₹x, Net Income will be decreased by ₹x. OR vice-versa but % may vary.

$$\text{e.g.}\quad \begin{array}{l} \overset{NI}{50} + \overset{T}{30} = \overset{I}{80} \\ \overset{-5}{45} + \overset{+5}{35} = 80 \end{array} \Big] \text{same}$$

TRICKY :- Income is same

$$19\% \text{ of Tax} = 6\% \text{ of Net Income}$$

$$\Rightarrow \frac{19}{6} = \frac{\text{Net Income}}{\text{Tax}} \Rightarrow \text{Income} = 19+6 = 25$$

$$\text{Tax \%} = \frac{\text{Tax}}{\text{Income}} \times 100 = \frac{6}{25} \times 100 = 24\%$$

Ans.

If income Tax is increased by 19%, Net income get decreased by 1%. Find the Tax Rate?

Solution :- 19% of Tax = 1% of Net INCOME

$$\Rightarrow \frac{\text{Tax}}{\text{Net Income}} = \frac{1}{19} \Rightarrow \text{Income} = 1+19 = 20$$

$$\text{Tax \%} = \frac{1}{20} \times 100 = 5\%$$

Ans.

A candidate got 25% marks and fail by 30 marks while another candidate who scores 50% marks gets 20 marks more than minimum required marks to pass the Examination. Find the maximum and passing marks for the Exam?

Solution :- equate passing marks

I $\quad\quad\quad$ II

$25\% + 30 = 50\% - 20$

$\Rightarrow 25\% = 50 \quad \therefore mm\,(100\%) = 2 \times 100 = 200$

$1\% = 2 \quad$ Passing marks $= 25\% + 30$

$= 80$ Ans

किसी परीक्षा में 60 % छात्र अंग्रेजी में उत्तीर्ण होते हैं। 70 % हिन्दी में और 40% दोनों विषयों में उत्तीर्ण (पास) होते हैं। तो दोनों में फेल होने वाले छात्रों का प्रतिशत क्या होगा?

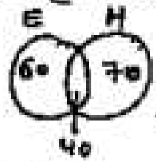
Solution :— I (FORMULA)

$N(A \cup B) = N(A) + N(B) - N(A \cap B)$

$= 70 + 60 - 40 = 90$

failed in both $= 100 - 90 = 10\%$

II (VENN Diagram)



$= 60 + 70 - 40$

$= 90\%$ pass

fail $= 100 - 90 = 10\%$

किसी परीक्षा में 80% छात्र अंग्रेजी में और 85% गणित में उत्तीर्ण होते हैं। जबकि 75% दोनों में उत्तीर्ण होते हैं। यदि 45 छात्र इन दोनों विषयों में अनुत्तीर्ण हुए हों तो कुल छात्रों की संख्या ज्ञात करो?

Solution :-

$E \rightarrow 80\%$
$H \rightarrow 85\%$] $+165\%$
$\quad -75$
$\quad 90\%$ — pass

fail $= 100 - 90 = 10\%$

$10\% \longrightarrow 45$

$100\% \longrightarrow 450$ Ans.

रामबाण (जिसको माना वो नहीं आएगा)
जो बचेगा वो आएगा

A total of ₹ 180 is to be divided among 100 students such that each girl student receives ₹2 and each boy student receives ₹ 1.5. Find the Number of girl students?

Solution :— I

let all are boys

money divided will be

$= 100 \times 1.5$

$= 150$

money Rem $\Rightarrow 180 - 150 = 30$

which was due to diff of .50

girls $= \dfrac{30}{0.5} = 60$ ANS.

(Boys माने तो girl आएगा)

OR

II

माना सारी लड़कियां थी 100 की 100

money $= 100 \times 2 = 200$

मिलना ज्यादा $= 200 - 180$
$= 20$

क्योंकि लड़कों को .5 कम मिल रहे थे।

लड़के $= \dfrac{20}{.5} = 40$

(girls माने तो boys आएगा।)

$\Rightarrow$ girls $= 100 - 40 = 60$

In a zoo, there are only pigeons and rabbits if there are total of 200 hedds and 560 legs then find the Number of pigeons and rabbits?

Solution :— Let all are pigeons (200)

$\therefore$ legs $= 200 \times 2 = 400$

extra $= 560 - 400 = 160$

but extra 160 is due to 2 more legs of Rabbit

Rabbit $= \dfrac{160}{2} = 80$ ANS

(Pigeon माना तो Rabbit आएगा।)

Pigeon $= 200 - 80 = 120$ ANS.

OR

let all are Rabbits (200)

$\therefore$ legs $= 200 \times 4 = 800$

gap $= 800 - 560 = 240$

240 ज्यादा coz pigeon has two less legs

pigon $= \dfrac{240}{2} = 120$

(Rabbit माना तो Pigeon आएगा)

Rabbit $= 200 - 120 = 80$

Each correct answer fetches 4 marks and each wrong answer fetches a penalty of 1 mark. If a student attempts 48 questions and scores 132 marks. then find the number of correct and wrong answers?

Solution:- let he attempted all (48) correct

Total marks $= 48 \times 4 = 192$

but his actual score is 132. Difference of 60 is due to diff. of $5(4-(-1))$ b/w correct and wrong

$\#$ wrong $= \frac{60}{5} = 12$ [correct माना तो wrong आएगा]

$\#$ correct $= 48 - 12 = 36$ Ans

Population of a town is 15000. If the number of the males ↑ by 8%. and that of females by 10%. then the population would increase to 16300 after 1yr. Find the number of females?

Solution:-

I

let all are male

∴ Population after 1yr

$= 15000 + \frac{15000 \times 8}{100}$

$= 16200$

gap $= 16300 - 16200$

$= 100$ less

$\#$ females $= \frac{100}{10\% - 8\%} \times 100 = 5000$

(male माना तो female आया)

II

let all are females

Population after 1yr

$= 15000 + \frac{15000 \times 10}{100}$

$= 16500$

gap $= 16500 - 16300$

$= 200$ Extra

$\#$ males $= \frac{200}{2\%} \times 100$

$= 10,000$

(female माना तो male आया)

$\#$ female $= 15000 - 10000 = 5000$

A man lent 2000 partly at 5% and balance at 4%. If he receives ₹ 92 as annual interest. find the amount lent at 5%?

Solution :- Let whole amnt is invested at 4%

SI after 1yr = $\frac{2000 \times 4}{100} = 80$ ₹

gap = 92 − 80 = 12 ₹ [coz Rate diff = 5% − 4% = 1%]

amnt lent at 5% = $\frac{12}{1} \times 100 = 1200$ **ANS.**

[4% माना तो 5% वाला आयेगा]

A man covers a total of 240 km in 6 hours partly at 30 km/hr and rest at 60 km/hr. what distance was covered at 30 km/hr.

Solution:- Note → Ratio of speed gives ratio of Time

let he covered all distance at 60 km/hr

∴ in 6 hours D = 60 × 6 = 360 km

gap = 360 − 240 = 120 km

Time for 30 km/hr Journey = $\frac{120}{60-30} = 4$ hr

[60 को माना तो 30 का Time आया]

D = 4 × 30 = 120 Km **ANS.**

PROFIT & LOSS

Profit, Loss & Discount
(लाभ, हानि और बट्टा)

Terminology :-

Cost Price (C.P.) क्रय मूल्य
The Price at which article is bought or money that goes out of pocket.

Selling Price (S.P.) विक्रय मूल्य
The price at which article is sold or the amount of money that comes into pocket.

Note :- The Selling Price (S.P.) of the seller is the cost Price of the buyer.

PROFIT (लाभ) if $SP > CP$

$$P = SP - CP = \text{वि० मू०} - \text{क्र० मू०}$$

LOSS (हानि) if $CP > SP$

$$L = CP - SP = \text{क्र० मू०} - \text{वि० मूल्य}$$

IF PROFIT on Selling an article for ₹ 425 is same as loss on selling it for ₹ 355. Find the cost price of the Article?

Solution :- I (BASIC)

Let C.P. = X

P = L

$425 - X = X - 355$

$\Rightarrow 2X = 780 \Rightarrow X = 390$ A.

II (TRICKY)

Find $SP_2 - SP_1 = 425 - 355$
$= 70$

70
1 : 1
35 35
P L

$\therefore CP = 425 - 35 = 390$

OR $CP = 355 + 35 = 390$

OR III $\frac{SP_1 + SP_2}{2} = \frac{425 + 355}{2}$
$= 390$ ANS.

किसी वस्तु को 1560 में बेचने पर जो लाभ होता है, वह उसी वस्तु को 760 में बेचने पर हुई हानि से 3 गुना है। तो उस वस्तु का क्र० मू० क्या होगा?

Solution :- I (BASIC)

Let C.P. = X

P = 3 L

$\therefore 1560 - X = 3(X - 760)$

$\Rightarrow 4X = 3840 \Rightarrow X = 960$ ANS.

II $1560 - 760 = 800$

3 : 1
600 200
P L

$CP = 1560 - 600 = 960$

OR $CP = 760 + 200 = 960$ Ans.

IF the Profit on selling an article for ₹ 480 is 25% more than loss on the selling it for ₹ 300. Find the cost Price of article?

Solution :-

P : L

125 : 100

5 : 4

$SP_2 - SP_1 = 480 - 300$
$= 180$

5 : 4
P 100 80 L

$\Rightarrow CP = 480 - 100 = 380$

OR $CP = 300 + 80 = 380$ ANS.

Profit and Loss is always Calculated/Reckoned on cost price unless otherwise stated.

लाभ और हानि प्रतिशत हमेशा क्रय मूल्य (C.P.) पर ज्ञात किए जाते हैं। परन्तु अगर Examiner बोलता है कि आपको वि०मू० पर निकालना है तो आप वि०मू० पर ही निकालें।

PROFIT % $= \dfrac{\text{PROFIT}}{\text{COST Price}} \times 100 = \dfrac{P}{CP} \times 100$

$$\text{LOSS \%} = \frac{\text{LOSS}}{\text{COST PRICE}} \times 100 = \frac{L}{CP} \times 100$$

e.g. CP = 800 SP = 1000

$$\therefore P\% = \frac{200}{800} \times 100 = 25\%$$

CP = 800, SP = 400

$$\therefore L\% = \frac{400}{800} \times 100 = 50\% \text{ loss}$$

PROFIT (P%)

$$SP = CP * \frac{(100 + P\%)}{100}$$

$$\text{OR } CP = \frac{SP * 100}{(100 + P\%)}$$

LOSS (L%)

$$SP = CP * \frac{(100 - L\%)}{100}$$

$$\text{OR } CP = \frac{SP * 100}{(100 - L\%)}$$

METHOD 1 (100% वाला तरीका)

आपको क्रमूल्य (CP) को हमेशा 100% लेना है।

e.g. let P% = 20% → CP = 100%, SP = 120%

L% = 20% → CP = 100%, SP = 80%

CP = 600, P% = 20%, SP = ?

100% → 600

1% → $\frac{600}{100}$

SP = 120% → $\frac{600}{100} \times 120 = 720$ ANS.

SP = 5120 L% = 20%, CP = ?

L% = 20% → CP = 100%, SP = 80% → 5120

1% → $\frac{5120}{80}$

CP = 100% → $\frac{5120}{80} \times 100$

6400 ANS.

CP = ? P% = 25% SP = 6250

P% = 25% → CP = 100%, SP = 125% → 6250

CP 100% → $\frac{6250}{125} \times 100 = 5000$

METHOD 2 (Ratio वाला तरीका)

इसमें सबसे पहले आपको P% या L% को fraction में convert करना है।

e.g. $P\% = 20\% = \frac{1 \to P}{5 \to CP} \Rightarrow SP \to 5+1 = 6$

$L\% = 20\% = \frac{1 \to L}{5 \to CP} \Rightarrow SP = 5-1 = 4$

$P\% = 12.5\% = \frac{1 \to P}{8 \to CP} \Rightarrow SP = 8+1 = 9$

$L\% = 16\frac{2}{3}\% = \frac{1 \to L}{6 \to CP} \Rightarrow SP = 6-1 = 5$

$P\% = 14\frac{2}{7}\%$, $CP = 490$

$SP = ?$

$14\frac{2}{7}\% = \frac{1 \to P}{7 \to CP} \Rightarrow SP = 8$

$7 \longrightarrow 490$

$1 \longrightarrow 70$

$8\,(SP) \longrightarrow 70 \times 8 = 560$ Ans.

$L\% = 10\%$ $SP = 7290$

$CP = ?$

$10\% = \frac{1 \to L}{10 \to CP}$

$\Rightarrow SP \to 10-1 = 9$

$9 \longrightarrow 7290$

$10 \longrightarrow \frac{7290}{9} \times 10$

$= 8100$ Ans.

$P\% = 40\%$, $CP = 3000$

$SP = ?$

$40\% = \frac{2 \to P}{5 \to CP} \Rightarrow SP = 5+2 = 7$

$5 \longrightarrow 3000$

$7 \longrightarrow \frac{3000}{5} \times 7 = 4200$ Ans.

$L\% = 8.33\%$

$SP = 1210$, $CP = ?$

$8.33\% = \frac{1 \to L}{12 \to CP}$

$SP \to 12-1 = 11$

$11 \longrightarrow 1210$

$12 \longrightarrow \frac{1210}{11} \times 12 = 1320$ Ans.

If Ratio of cost price and selling price is 8:9. Find Profit/Loss %?

Solution :- $\frac{CP}{SP} = \frac{8}{9} \Rightarrow \text{Profit} = 1 \Rightarrow P\% = \frac{1}{8} \times 100$

$= 12.5\%$

If SP is $\frac{5}{4}$ of CP. Find Profit %?

Solution :- SP : CP

$\frac{5}{4} : 1 \Rightarrow P\% = \frac{1}{4} \times 100$

OR 5 : 4

$= 25\%$ ANS.

If Profit is calculated on SP then P% = 20%. what is original P% (on CP)?

Solution :- $20\% = \frac{1 \rightarrow P}{5 \rightarrow SP} \Rightarrow CP = 4$

Actual P% on CP $= \frac{1}{4} \times 100 = 25\%$ ANS

If Loss% is calculated on SP, then L% = 10%. what is original Loss%? (on CP)

Solution :- $L\% = 10\% = \frac{1 \rightarrow L}{10 \rightarrow SP} \Rightarrow CP = 10 + 1 = 11$

Actual L% $= \frac{1}{11} \times 100 = 9.09\%$ ANS

P% on CP = 12.5% then P% on SP?

$12.5\% = \frac{1 \rightarrow P}{8 \rightarrow CP} \Rightarrow SP \rightarrow 9$

P% ON SP $= \frac{1}{9} \times 100 = 11.11\%$ ANS.

By selling a mobile for ₹ 1045 a man loss 5%. At what price should he sell the mobile to gain 5%?

Solution :- I (Basic)

CP = 100% always

SP_1 = 95% [5% Loss]

SP_2 = 105% [5% gain]

95% ⟶ 1045

⇒ 105% ⟶ $\frac{1045}{95} \times 105 = 1155$

II (Tricky)

5% Loss = $\frac{1 \to L}{20 \to CP}$

$SP_1 = 19$

5% gain = $\frac{1 - P}{20 - CP}$

SP_1 SP_2 — 21

19   ×55

×55

1045   1155

एक आदमी को एक पैंट को 1920 में बेचने पर 20% की हानि हुई। यदि वह उसे 2520 ₹ में बेचता तो कितने % की लाभ या हानि होती?

Solution :- I (BASIC)

1920 ⟶ 80% [20% L]

1 ⟶ $\frac{80\%}{1920}$

2520 ⟶ $\frac{80\%}{1920} \times 2520$

= 105%

105% means 5% Profit

II (TRICKY)

$$\frac{SP_1}{100 \pm P/L} = \frac{SP_2}{100 \pm P/L}$$

$$\frac{192}{80} = \frac{252}{100 \pm P/L}$$

⇒ $100 \pm P/L = 105$

⇒ $\pm P/L = 5$

⇒ P% = 5%

An Article is sold at 8% loss. Had the shopkeeper sold it for ₹ 650 more he would have gain 5%. Find the cost price?

Solution :- I (BASIC)

Let CP = ₹ X

SP₁ SP₂

$X[.92] + 650 = X[1.05]$

$\Rightarrow X[.13] = 650$

$\Rightarrow X = 5000$ ANS.

II (TRICKY)

$\boxed{100\%} \rightarrow CP$

SP₁ → 92% SP₂ → 105%

$13\% \rightarrow 650$

$\therefore 100\% \rightarrow 5000$ ANS

एक दुकानदार ने एक मोबाईल 5% हानि पर बेचा। अगर वो पहले से इसको ₹ 3375 ज्यादा में बेचता तो उसको अब 10% का लाभ होता। वह उस मोबाईल को कितने ₹ में बेचे कि अब उसको 20% का लाभ हो?

Solution :- I (BASIC)

माना CP = ₹ X

SP₁ SP₂

$X[.95] + 3375 = X[1.10]$

$\Rightarrow X[.15] = 3375$

$\Rightarrow X = 22500 \rightarrow CP$

$SP_3 = 22500[1.20]$

$= 27000$ ₹ ANS.

II (TRICKY)

$\boxed{100\%} \rightarrow CP$

SP₁ → 95% SP₂ → 110%

15%

$15\% \rightarrow 3375$

$SP_3\ [120\%] \rightarrow \frac{3375}{15} \times 120$

$= ₹27000$ (4)

\# A shopkeeper sells an article at 12% Profit. Had he bought it at a price 10% Less than earlier and sold for ₹ 250 more. he would have gain 30%. Find the cost price of Article?

Solution :– I (BASIC)

Let CP = X ₹

$$\underset{\downarrow}{SP_1} \qquad\qquad \underset{\downarrow}{SP_2}$$

$$X[1.12] + 250 = X[.90][1.30]$$

$$\Rightarrow X[1.12] + 250 = X[1.17]$$

$$\Rightarrow X[.05] = 250 \Rightarrow X = 5000$$

II (TRICKY)

	old	New
let CP =	100	90
	↓ 12% P	↓ 30% P
SP →	112	117

+5

$$5 \longrightarrow 250$$

$$\therefore CP(100) \longrightarrow 5000 \text{ ANS.}$$

\# एक दुकानदार एक वस्तु को 20% लाभ पर बेचता है। अगर वह उस वस्तु को 20% पहले से कम पर खरीदे और ₹ 75 कम पर बेचे तो उसे अब 25% का लाभ होता है। तो उस वस्तु का क्रय मूल्य बताओ?

Solution :– I (BASIC)

माना CP = ₹ X

$$\underset{\downarrow}{SP_1} \qquad\qquad \underset{\downarrow}{SP_2}$$

$$X[1.20] - 75 = X[.80][1.25]$$

$$\Rightarrow X[1.20] - 75 = X[1]$$

$$\Rightarrow X[.20] = 75$$

$$X = ₹\ 375 \text{ ANS.}$$

II (TRICKY)

	old	−20%	NEW
CP =	100		80
	20% ↓ P		25% ↓ P
SP =	120		100

−20

$$20 \longrightarrow 75$$

$$CP = 100 \longrightarrow 75 \times 5$$

375 ₹ ANS

\# A shopkeeper sells his goods at 25% Profit. Had he purchased it for ₹ 900 less and sold for ₹ 900 less he would have gain 5% more. Find the original cost price?

Solution :- I (BASIC) II III (TRICKY)

$\underset{SP}{125\%} - 900 = \underbrace{(\underset{CP}{100\%} - 900)(1.30)}_{\text{New SP}}$

$1250\% - 9000 = 1300\% - 11700$

$\Rightarrow 50\% = 2700$

$\therefore 100\% = CP = 5400$ ANS.

$\text{TRICK} = \dfrac{\text{New Profit}}{P_2 - P_1} \times \text{Gap}$

$= \dfrac{30\%}{5\%} \times 900$

$= 5400$ ANS.

\# एक दुकानदार अपनी वस्तु को 20% लाभ पर बेचता है। यदि वह क्रय मूल्य और विक्रय मूल्य दोनों को 100₹ कम कर दे तो उसका लाभ 4% बढ़ जाता है। तो वस्तु का क्र० मू० बताओ?

Solution :- I (BASIC) II TRICKY

$20\%\ P = \dfrac{1 \rightarrow P}{5 - CP}$ $SP - 6$

$CP_1 : SP_1 = 5 : 6$

$24\% = \dfrac{6 \rightarrow P}{25 - CP}$ $SP \rightarrow 31$

$CP_2 : SP_2 = 25 : 31$

$\therefore \dfrac{5x - 100}{6x - 100} = \dfrac{25}{31}$

$\Rightarrow 5x = 600$ ANS.

CP : SP

$5_{\times 6} \overset{1}{:} 6_{\times 6}$ (5)

$25_{\times 1} \underset{6}{:} 31_{\times 1}$ (5)

5 unit → 100

30 unit CP → 600 ANS

A shopkeeper sells his goods at a profit of 20%. If he reduces both CP and SP by ₹10 and ₹5 respectively then the profit % ↑ by 10%. Find the cost price of article?

Solution :- I

$20\%\ P = \frac{1 - P}{5 - CP} \rightarrow SP - 6$

let $CP = 5x,\ SP = 6x$

Now $30\%\ P = \frac{3 - P}{10 - CP} \rightarrow SP - 13$

$\therefore \frac{5x - 10}{6x - 5} = \frac{10}{13}$

OR $65x - 130 = 60x - 50$

$\Rightarrow 5x = 80 = CP$ ANS

II

$(120\% - 5) = (100\% - 10)(130)$

OR $1200\% - 50 = 1300\% - 130$

$\Rightarrow 100\% = 80$ ANS.

एक दुकानदार एक वस्तु को 25% के लाभ पर बेचता है। यदि वह अपने क्र० मू० और वि० मू० को क्रमशः 20 ₹ और 4 ₹ बढ़ा दे तो उसका लाभ 15% घट जाता है। तो वस्तु का क्रय मूल्य ज्ञात करो?

Solution :-

$$(125\% + 4) = (100\% + 20) \times \frac{110}{100}$$

$\Rightarrow 1250\% + 40 = 1100\% + 220$

$150\% = 180 \quad \therefore 100\% = CP = \frac{180}{150} \times 100$

$= 120$ ₹ ANS.

ARTICLE BASED QUESTIONS

Note :- In this type of Question आपको Articles (वस्तुएँ) को बराबर रखना होगा।
e.g. if you have 20 articles CP then you must have 20 article SP

Articles were bought at 6 for ₹ 5 and sold for 5 for ₹ 6. Gain/loss percent is ?

Solution :- I (Basic)
make the Article equal at Lcm of 6, 5 = 30

	A	P	
Buy	5×6	$5 \times 5 = 25$	CP
Sold	6×5	$6 \times 6 = 36$	SP

gain % $= \frac{11}{25} \times 100 = 44\%$ Ans.

II (Tricky)

A	P	
6	5	CP
5	6	SP

CP = 25
SP = 36
P% $= \frac{11}{25} \times 100 = 44\%$.

एक दुकानदार कुछ वस्तुओं को 15 ₹ में 12 वस्तुओं की दर से खरीदता है और सभी वस्तुओं को 14₹ में 10 वस्तुओं के हिसाब से बेच देता है। तो कितने प्रतिशत लाभ या हानि हुई ?

Solution :-

	A	P		
Buy	12	15	→ 150	→ CP
Sold	10	14	→ 168	→ SP

$\Rightarrow P\% = \frac{18}{150} \times 100$
$= 12\%$ Ans.

IF COST PRICE OF 18 articles is equal to selling price of 15 articles. The gain % is?

Solution :-

I

let CP of 18 = SP of 15 = 90

$\Rightarrow$ CP of 1 article = 5

$\Rightarrow$ SP of 1 article = 6

gain % = $\frac{1}{5}$ = 20%

II

CP of 18 = SP of 15

$\Rightarrow \frac{CP}{SP} = \frac{15}{18} = \frac{5}{6}$

$\Rightarrow$ gain % $= \frac{1}{5} \times 100$

$= 20\%$

अगर 9 वस्तुओं का क्रय मूल्य 12 वस्तुओं के वि० मू० के बराबर हो तो लाभ/हानि % बताएं?

Solution :

I

माना CP of 9 = SP of 12 = 36

CP of 1 article = 4

SP of 1 article = 3

loss % = $\frac{1}{4} \times 100 = 25\%$

II

CP of 9 = SP of 12

$\Rightarrow \frac{CP}{SP} = \frac{12}{9} = \frac{4}{3}$

$\Rightarrow$ loss % $= \frac{1}{4} \times 100$

$= 25\%$

IIIRD METHOD $= \frac{\text{\# Good LEFT}}{\text{\# Goods sold}} \times 100$

CP of 18 = SP of 15

% Profit $= \frac{3}{15} \times 100$

$= 20\%$ ANS.

CP of 9 = SP of 12

% LOSS $= \frac{3}{12} \times 100$

$= 25\%$ ANS.

इस Type के questions में आपको focus करना है कि Profit/Loss SP या CP किसके रूप में given है। अगर PROFIT/LOSS SP के रूप में given है तो 1 article का SP = 1₹ मान लो और अगर Profit/loss CP के रूप में given है तो 1 article का CP = 1₹ मान लो।

e.g Profit = SP of 11 articles ⇒ let SP of 1 article = 1₹

Loss = CP of 11 articles ⇒ let CP of 1 = 1₹

By selling 33m of cloth a shopkeeper gains CP of 11m. find gain % ?

Solution :- I

gain = SP − CP

∴ SP of 33m − CP of 33m = CP of 11m

⇒ SP of 33m = CP of 44m

$\Rightarrow \frac{CP}{SP} = \frac{33}{44}$

$\Rightarrow$ gain % $= \frac{11}{33} \times 100 = 33\frac{1}{3}\%$

II

let CP of 1m = 1₹

⇒ CP of 33m = 33₹

gain = CP of 11m = 11₹

gain % $= \frac{11}{33} \times 100$

$= 33.33\%$ An

By Selling 33m of cloth a shopkeeper gains SP of 11m. find gain % ?

Solution :- I

SP of 33m − CP of 33m = SP of 11m

⇒ SP of 22m = CP of 33m

$\Rightarrow \frac{CP}{SP} = \frac{22}{33} \Rightarrow$ gain % $= \frac{11}{22} \times 100$

$= 50\%$

II

let SP of 1m = 1₹

⇒ SP of 33m = 33₹

gain = SP of 11m = 11₹

CP of 11m = 33 − 11 = 22

gain % $= \frac{11}{22} \times 100$

$= 50\%$

By selling 33m of cloth a shopkeeper loss CP of 1m. find loss%?

Solution :- I

Loss = CP − SP

CP of 33m − SP of 33m

= CP of 11m

$\Rightarrow$ CP of 22m = SP of 33m

$\Rightarrow \frac{CP}{SP} = \frac{33}{22}$

loss % = $\frac{11}{33} \times 100 = 33\frac{1}{3}\%$

II

let CP of 1m = 1 ₹

CP of 33m = 33 ₹

loss = CP of 11m = 11 ₹

loss % = $\frac{11}{33} \times 100$

= $33\frac{1}{3}\%$ Ans

By selling 33m of cloth a shopkeeper losses SP of 11m. find loss%?

Solution :- I

CP of 33m − SP of 33m

= SP of 11m

$\Rightarrow$ CP of 33m = SP of 44m

$\Rightarrow \frac{CP}{SP} = \frac{44}{33}$

loss % = $11/44 \times 100 = 25\%$

II

let SP of 1m = 1 ₹

SP of 33m = 33 ₹

loss = SP of 11m = 11 ₹

$\Rightarrow$ CP of 33m = 33 + 11

= 44

$\therefore$ loss % = $\frac{11}{44} \times 100 = 25\%$

36 संतरे बेचने पर एक दुकानदार को 4 संतरों के क्र० मू० के बराबर लाभ होता है। तो लाभ %?

Solution :- I

SP of 36 − CP of 36

= CP of 4

$\Rightarrow$ SP of 36 = CP of 40

$\Rightarrow \frac{SP}{SP} = \frac{36}{40} = \frac{9}{10}$

$\Rightarrow$ P% = $\frac{1}{9} \times 100 = 11\frac{1}{9}\%$

II

माना 1 संतरे का क्र०मू० = 1 ₹

लाभ = 4 संतरे का क्र०मू० = 4

लाभ % = $\frac{4}{36} \times 100$

= $11\frac{1}{9}\%$ Ans.

36 संतरे बेचने पर एक दुकानदार को 4 संतरों के विक्रय मूल्य के बराबर लाभ होता है। लाभ % क्या होगा?

Solution :- I (Basic) OR II (Tricky)

I (Basic)

SP of 36 − CP of 36
= SP of 4
⇒ SP of 32 = CP of 36
OR $\frac{CP}{SP} = \frac{32}{36} = \frac{8}{9}$

लाभ % $= \frac{1}{8} \times 100 = 12.5\%$

II (Tricky)

माना 1 संतरे का वि० मू० = 1 ₹
लाभ = 4 संतरों का विक्रमू० = 4
36 संतरों का वि० मू० = 36
⇒ 36 संतरों का क्र० मू० = 36 − 4 = 32

∴ लाभ % $= \frac{4}{32} \times 100 = 12.5$

36 संतरे बेचने पर एक दुकानदार को 4 संतरों के क्र०मू० के बराबर हानि होती है। हानि प्रतिशत क्या होगा?

Solution :- I (Basic)

CP of 36 − SP of 36
= CP of 4
⇒ CP of 32 = SP of 36
⇒ $\frac{CP}{SP} = \frac{36}{32} = \frac{9}{8}$

LOSS % $= \frac{1}{9} \times 100 = 11\frac{1}{9}\%$

II (Tricky)

माना 1 संतरे का CP = 1 ₹
हानि = 4 संतरों का CP = 4 ₹
36 संतरों का CP = 36 ₹
हानि % $= \frac{4}{36} \times 100 = 11\frac{1}{9}\%$

36 संतरे बेचने पर एक दुकानदार को 4 संतरों के विक्रय मूल्यों के बराबर हानि होती है। हानि % क्या होगा?

Solution :- I (BASIC)

CP of 36 − SP of 36
= SP of 4
⇒ CP of 36 = SP of 40
OR $\frac{CP}{SP} = \frac{40}{36}$

∴ हानि % $= \frac{4}{40} \times 100$
$= 10\%$ ANS.

II (TRICKY)

माना 1 संतरे का विक्रमू० = 1 ₹
हानि = 4 संतरों का SP = 4
36 संतरों का SP = 36
⇒ 36 संतरों का CP = 36 + 4 = 40
हानि % $= \frac{4}{40} \times 100 = 10\%$

A man buys some apples at the rate of 1 Apple for ₹ 2 and equal no. of apples at the rate of 2 apple for ₹ 1. He sells all of them @ 4 apples for ₹ 3 find the profit/loss % ?

Solution :–

	Article	Price
ये दोनों same होने चाहिए	2×1	$2 ₹ \times 2 = 4$
	2	$1 ₹ = 1$

CP of 4 $= 4+1 = ₹ 5$

SP of 4 $= ₹ 3$

$\Rightarrow$ loss % $= \frac{2}{5} \times 100 = 40\%$ Ans

1 दुकानदार कुछ वस्तुओं को 4 ₹ में 3 वस्तुओं की दर से खरीदता है और वह उतनी ही वस्तु 5 ₹ में 4 वस्तु की दर से खरीदता है। इन सभी वस्तुओं को 3 ₹ में 2 वस्तु की दर से बेच देता है। उसे कितने % Profit या loss हुआ ?

Solution :–

Article	Price
4×3	$4 \times 4 = 16$
3×4	$5 \times 3 = 15$

24 वस्तुओं का CP $= 31$

2 वस्तुओं का SP $= 3$ ₹

24 वस्तुओं का SP $= 3 \times 12 = 36$

Profit % $= \frac{5}{31} \times 100 = \frac{500}{31}\%$

A shopkeeper purchases oranges 30 for ₹ 100. How many should he sell in ₹ 100 to gain 20%?

Solution :- I (Basic)

CP of 30 oranges = 100

SP of 30 " $= 100 \times 1.20$

$= 120$ ₹

Now 120 ₹ में — 30 orange

$\Rightarrow$ 100 ₹ में — $\frac{30}{120} \times 100$

— 25 orange Ans.

II (Tricky)

$E = P \times Q$

$E = 100 = $ same

$\Rightarrow P \propto \frac{1}{Q}$

	old	NEW
Price	100	120
Qty	120	100
	6 :	5

$\times 5$: 6 → 30, 5 → 25 Ans.

एक दुकानदार ने 1 ₹ में 25 टॉफियां खरीदी। वह 1 ₹ में कितनी टॉफियां बेचे कि उसे 25% का लाभ हो?

Solution :- I (BASIC)

25 टॉफियों का क्र० मू० = 1

25 टॉफियों का वि० मू० = 1.25

$\Rightarrow$ 1.25 ₹ में बेचें 25

$\Rightarrow$ 1 ₹ में बेचें $\frac{25}{1.25} = 20$

II (TRICKY)

E = same = 1 ₹

$P \propto \frac{1}{Qty}$

	old	New
Price	100	125
Qty	125	100
	5 :	4

$5 \times$: 5 → 25, 4 → 20 Ans. ($\times 5$)

Note :→ अगर Sale / Purchase की Price Same है तो Price और Qty का Ratio एक दूसरे का उल्टा होगा। अगर नहीं Same है तो starting में Same रख लेंगे और बाद में Unitary method का इस्तेमाल करेंगे।

By selling 15 oranges for 1 ₹ a man loss 20%. How many for 1 ₹ should he sell to gain 20%?

Solution :- I (Basic)

SP_1 of 15 oranges = 1 ₹

CP of 15 oranges = $\frac{1}{.80} = \frac{5}{4}$

SP_2 of 15 oranges = $\frac{5}{4} \times 1.20$

= 1.5 ₹

∴ 1.5 ₹ में बेचे = 15

तो 1 ₹ में बेचे = $\frac{15}{1.5}$ = 10

II (Tricky)

Exp = Price × Qty

E = Same = 1 ₹

Price ∝ $\frac{1}{Qty}$

	old	New
Price	80	120
Qty	120	80

$5 \times (3 : 2) \times 5$

15 : 10 ANS.

By selling 45 oranges for ₹ 40 a man loss 20%. How many should he sell for ₹ 24 to earn 20%?

Solution :- I (Basic)

SP_1 of 45 oranges = 40

CP of 45 " = $\frac{40}{.80} = 50$

SP_2 of 45 oranges = 50 × 1.20

= 60 ₹

∴ 60 ₹ में बेचे 45 संतरे

तो 1 ₹ में बेचे $\frac{45}{60}$

और 24 ₹ में बेचे = $\frac{45}{60} \times 24$

= 18

II (Tricky)

Sale Price ≠ Purchase/Sale

40 24

[but हम पहले 40 का निकाल लेंगे (same रेट के लिए) बाद में 24 का unitary से]

	old	New
Price	80	120
Qty	120	80

$15 \times (3 : 2) \times 15$

45 : 30

40 ₹ में — 30

24 ₹ में — $\frac{30}{40} \times 24$ = 18

MARKED PRICE / LIST PRICE / TAG PRICE / LABELLED PRICE (अंकित मूल्य) M.R.P / Print Price

Price that is printed on the tag of article

Discount (बट्टा / छूट)

Discount = MP − SP = अं॰ मू॰ − वि॰ मू॰

Discount % हमेशा अंकित मू॰ पर ज्ञात किया जाता है।

D% is always calculated/Reckoned ON MP unless otherwise stated.

$$D\% = \frac{D}{MP} \times 100$$

Selling Price (S.P.) = $M.P. \times \frac{(100 - D\%)}{100}$

OR $M.P. = SP \times \frac{100}{(100 - D\%)}$

Note :- discount is always given by shopkeeper.

and if no discount ⇒ $MP = SP$

MARK UP %

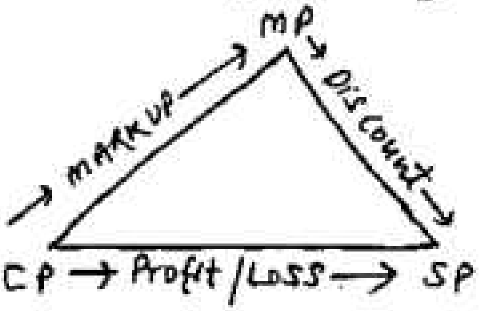
MARK UP = MP − CP

Note:- MARK UP % is always calculated on cost price.

$$\text{MARK UP \%} = \frac{\text{MARK UP}}{CP} \times 100$$

e.g CP = 100, MP = 150 ⇒ MARK UP % = 50%

Relation ship b/w CP, SP & MP



let P% = 20% and D = 25%

∴ $CP * (1.20) = SP$ — (1)

also $MP * (.75) = SP$ — (2)

from (1) and (2), we get

$$CP(1.20) = MP(.75)$$

100% METHOD [TAKE MP = 100%]

$D = 20\% \rightarrow \boxed{MP = 100\%}$, $\boxed{SP = 80\%}$

e.g. SP = ₹ 720, D% = 10% MP = ?

$D = 10\% \Rightarrow MP = 100\%$

$SP = 90\%$

$90\% \longrightarrow 720$

$\Rightarrow 100\% \longrightarrow \frac{720}{90} \times 100 = 800$ ANS.

Ratio METHOD

$D = 10\% = \frac{1 \rightarrow \text{discount}}{10 \rightarrow \text{MRP}} \Rightarrow SP = 10 - 1 = 9$

e.g. D = 10%, SP = 720, MP = ?

$D = 10\% = \frac{1 \rightarrow D}{10 \rightarrow MP} \Rightarrow SP = 10 - 1 = 9$

9 units $\longrightarrow 720$

MP = 10 units $\longrightarrow \frac{720}{9} \times 10 = 800$ ANS.

D = 12.5% MP = 1600, SP = ?

$D = 12.5\% = \frac{1 \rightarrow D}{8 \rightarrow MP} \Rightarrow SP = 7$

8 units $\longrightarrow 1600$

SP = 7 units $\longrightarrow 1400$ ANS.

DISCOUNT

I → SUCCESSIVE DISCOUNT (50% + 40%)

II → MP SP, CP (SP will be same)

III → ARTICLE BASED (BUY 2 get 1 Free)

Find the equivalent discount for 50% + 40%?

Solution :- I

$x + y + \frac{xy}{100}$

$x = -50,\ y = -40$

$= -50 - 40 + \frac{2000}{100}$

$= -70$ 70% ANS.

II

Let MP = 100

$100 \xrightarrow{-50\%} 50 \xrightarrow{-40\%} 30$

70%

III

$50\% = -\frac{1}{2}$

$40\% = -\frac{2}{5}$

old	New
2	1
5	3
10	3

70% D

If two successive discount of 60% + x% equivalent discount is 72%. Find x?

Solution → I

$-72 = -60 - x + \frac{60x}{100}$

$x = 30\%$

II

MP

$\boxed{100} \xrightarrow{60\%\,D} \boxed{40}$

$\boxed{100} \xrightarrow{-72\%} 28$

$\frac{12}{40} \times 100$

$= 30\%$ Ans

which offer is better (i) 60% + 30%
(ii) 70% + 20%

Solution → Total is same = 60 + 30 = 70 + 20 = 90

जिसकी numericle value बड़ी होगी वही ज्यादा discount होगा। (ii) is better coz of 70%

एक दुकानदार अंकित मूल्य पर तीन क्रमागत/लगातार बट्टा देता है। पहला discount 50%, दूसरा 10% और तीसरा x% है और तीनों का equivalent (समतुल्य) बट्टा 73% है तो x का मान बताईए?

Solution :- I (Basic)

$$SP = mP\left(\frac{100-D_1}{100}\right)\left(\frac{100-D_2}{100}\right)$$

$$27 = 100 \times \frac{50}{100} \times \frac{90}{100} \times \frac{100-D_3}{100}$$

$\Rightarrow D_3 = 40\%$ Ans.

II (Tricky)

mp $\boxed{100} \xrightarrow{50\%} \boxed{50} \xrightarrow{10\%} \boxed{45}$

$\boxed{100} \xrightarrow{73\%} \boxed{27}$, $\boxed{45} \rightarrow \boxed{27}$

$$\frac{18}{45} \times 100 = 40\%$$

marked price of an Article is 800 ₹. If after giving two successive discount selling price is 576. If 1st discount is 20%. Find second?

Solution :- I

$$576 = 800 \times \frac{80}{100} \times \frac{100-D}{100}$$

$\Rightarrow D = 10\%$ Ans

II (TRICKY)

mp $\boxed{800} \xrightarrow{20\%\ D} \boxed{640}$

$\boxed{640} \xrightarrow{D=64}$ SP $\boxed{576}$, $\boxed{800} \rightarrow$ SP $\boxed{576}$

$$D\% = \frac{64}{640} \times 100 = 10\% \text{ Ans.}$$

किसी वस्तु का अंकित मूल्य (mp) 1500 ₹ है। एक दुकानदार अंकित मूल्य पर दो लगातार बट्टा देता है। यदि एक बट्टा 30% है और दूसरा x% है। दुकानदार वस्तु को अन्त में 840 में बेचता है। x का मान क्या होगा?

Solution :- mp $\boxed{1500} \xrightarrow[450]{30\%} \boxed{1050}$

$\boxed{1050} \xrightarrow{D=210} \boxed{840}$, $\boxed{1500} \rightarrow \boxed{840}$

$$D\% = \frac{210}{1050} \times 100 = 20\% \text{ उत्तर}$$

CONCEPT / TRICK [SP = same]

$$SP = CP * \left[\frac{100 \pm P/L\%}{100}\right] = MP * \left[\frac{100 - D\%}{100}\right]$$

$$\therefore \; CP\left[\frac{100 \pm P/L\%}{\cancel{100}}\right] = MP * \left[\frac{100 - D\%}{\cancel{100}}\right]$$

$$\rightarrow \boxed{\begin{array}{ccc} CP & : & MP \\ 100 - D\% & : & 100 \pm P/L\% \end{array}}$$

e.g. $P\% = 20\%$, $D = 10\%$

$CP : MP = 90 : 120$
$= 3 : 4$

A shopkeeper after allowing a discount of 10% still gains 20%. If marked price is 800 find cost price of article?

Solution :-

$CP : MP$
$100 - D\% : 100 + P\%$
$90\% : 120\%$
$3 : 4$

$4 \longrightarrow 800$
$1 \longrightarrow \frac{800}{4}$
$3 = CP \rightarrow \frac{800}{4} \times 3$
$= 600$ ₹ ANS.

एक दुकानदार अगर 20% की छूट दे तो उसे 10% की हानि होती है। IF CP = 640, MP = ?

Solution :- I

$CP : SP : MP$
$10 \times 4 \quad 9 \times 4$
$\qquad 4 \times 9 \quad 5 \times 9$
$\overline{40} \qquad\qquad \overline{45}$

$16\left(\begin{array}{c} 40 \\ 1 \\ 640 \end{array}\right) \qquad 45 \Big) \times 16$

720 ANS.

II

$CP : MP$
$100 - D : 100 - L\%$
$80 : 90$
$8 : 9$

$80\left(1 \qquad 9\right) \times 80$
$640 \qquad 720$ ANS.

\# एक दुकानदार को अपनी वस्तुओं के ऊपर कितना % अधिक अंकित करना पड़ेगा कि 20% छूट देने के बाद भी 10% का लाभ हो ?

Solution :- I

$CP : MP$

$\boxed{100 - D\% : 100 + P\%}$

$100 - 20 : 100 + 10$

$80 : 110$

$\frac{30}{80} \times 100 = 37.5\%$

II

$P\% = 10\% = \frac{1}{10} \Rightarrow \frac{CP}{SP} = \frac{10}{11}$

$D\% = 20\% = \frac{1}{5} \Rightarrow \frac{SP}{MP} = \frac{4}{5}$

$CP : SP : MP$

$10 \times 4 \quad 11 \times 4$

$\quad\quad 4 \times 11 \quad 5 \times 11$

$40 : 55$

$= \frac{15}{40} \times 100 = 37\frac{1}{2}\%$

III Direct $\boxed{P + D \times \frac{100}{100 - D}}$

$= 30 \times \frac{100}{80} = 37.5\%$

\# By how much percentage a shopkeeper marks his goods above its CP so that after allowing 10% discount he may gain 30% P

Solution :- I

$CP : MP$

$100 - D\% : 100 + P\%$

$100 - 10 : 100 + 30$

$90 : 130$

$\frac{40}{90} \times 100 = 44\frac{4}{9}\%$

II (DIRECT)

$(P + D) * \frac{100}{100 - D}$

$= 40 * \frac{100}{90}$

$= 44\frac{4}{9}\%$

By selling an article for ₹ 750 & allowing 20% discount a shopkeeper still earns 40% profit. If the article is sold at 0% discount then profit would be?

Solution :– CP : MP
$100-20 : 100+40$
$80 : 140$
$4 : 7$

No discount [MP = SP]
CP : SP
$4 : 7$
$P\% = \frac{3}{4} \times 100 = 75\%$

एक दुकानदार अपनी वस्तु को अंकित मूल्य पर 15% छूट देकर 425 ₹ में बेचता है। यदि वह बट्टा ना दे तो उसे 25% का लाभ होता है। उस वस्तु का क्रय मूल्य बताओ?

Solution :– 85% (SP) ⟶ 425
100% (MP) ⟶ $425/85 \times 100 = 500$ ₹

IF No discount MP = SP ⇒ SP = 500

$P\% = 25\% \Rightarrow CP = 500 \times \frac{100}{125} = 400$ ₹ Ans.

एक दुकानदार अपनी वस्तु को 10% छूट देने के बाद भी 20% लाभ कमाता है। यदि वह छूट/बट्टे को बदलकर 20% कर दे तो उसे अब कितने % का लाभ होगा?

Solution :– $CP : MP = (100-10) : (100+20)$
$= 3 : 4$

Now discount 20% CP : SP $3 : 4 \times \frac{80}{100} = 15 : 16$

$P\% = \frac{1}{15} \times 100 = 6.66\%$ Ans

\# On Purchase of 3 shirts, 1 shirt is given free. Find the discount %?

Solution:– Let price of 1 shirt = 1 ₹

MP of 4 shirts = 4 [∵ 1 is given free.]

SP of 4 shirts = 3

$D\% = \frac{1}{4} \times 100 = 25\%$ Ans.

\# 6 पुस्तकें खरीदने पर 4 पुस्तक फ्री दी जाती हैं। तो बट्टा (discount) प्रतिशत क्या होगा।

Solution:– माना एक पुस्तक का मूल्य = 1 ₹

10 पुस्तकों का अंकित मूल्य = 10 ₹ [क्योंकि 4 तो फ्री है]

10 पुस्तकों का विक्रय मूल्य = 6

$D\% = \frac{4}{10} \times 100 = 40\%$ Answer

\# 4 पेन खरीदने पर एक पेन फ्री दिया जाता हैं और साथ में 20% अतिरिक्त छूट (discount) दिया जाता हैं। तो समतुल्य बट्टा (equivalent discount) क्या होगा?

Solution:– माना एक वस्तु का मूल्य = 100 ₹

5 वस्तुओं का अंकित मूल्य = 500 ₹

5 वस्तुओं का विक्रय मूल्य = 4 × 100 = 400 ₹

20% Extra Discount ∴ नया विक्रय SP $= 400 \times \frac{80}{100} = 320$

$D\% = \frac{500-320}{500} \times 100 = 36\%$ Ans.

एक दुकानदार अपनी वस्तु के अंकित मूल्य पर 4% की छूट (discount) देता है। और 15 वस्तुएं खरीदने पर 1 वस्तु फ्री भी देता है। फिर भी उसको 35% का लाभ होता है। तो दुकानदार ने वस्तु का क्र०मू० अंकित मूल्य से कितने प्रतिशत अधिक रखा था?

Solution :- I (BASIC)

16 वस्तुओं का अंकित मूल्य = 1600

16 " " वि० मू० = 1500

4% D extra

नया वि० मू० $= 1500 \times \frac{96}{100}$

$= 1440$ ₹

35% लाभ

135% ⟶ 1440

100% (CP) ⟶ $\frac{1440}{135} \times 100 = \frac{3200}{3}$

CP : MP

$\frac{3200}{3}$: 1600 ⇒ CP : MP

2 : 3

50% ANS.

II (TRICKY)

CP : MP

100−D% : 100+P%

100−4% : 100+35

$\frac{96}{16}$: $\frac{135}{15}$

6 : 9

2 : 3

MARKUP% $= \frac{1}{2} \times 100$

$= 50\%$

A shopkeeper gives 3 articles free on purchase of 5 articles. He also allows a discount of 20% & still earns 25% profit. Find the MARKUP %?

Solution:- I (BASIC)

CP : MP

$5 \times \frac{80}{100} \times \frac{100}{125}$: 8

32 : 80

150% ANS.

II (TRICKY)

CP : MP

100−20 : 100+25

$\frac{80}{8}$: $\frac{125}{5}$

10 : 25

150%

सुमन एक दुकानदार से 18₹ प्रति दर्जन की दर से 6 पेंसिल खरीदता है। और दुकानदार सुमन को 1 पेंसिल मुफ्त में देता है। तो सुमन का लाभ प्रतिशत बताओ?

Solution :— I (BASIC)

12 पेंसिल → 18₹

6 पेंसिल → 9₹

प्राप्त पेंसिल → 7

7 पेंसिल → $\frac{18}{12} \times 7 = \frac{21}{2}$

$\frac{3}{2}$ लाभ

लाभ % $= \frac{3/2}{9} \times 100 = 16\frac{2}{3}$ %

II (TRICK)

लाभ % $= \frac{\text{मुफ्त वस्तु}}{\text{खरीदी गई वस्तु}} \times 100$

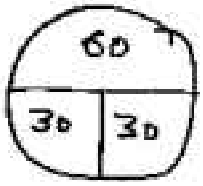
$= \frac{1}{6} \times 100$

$= 16\frac{2}{3}$ %

एक दुकानदार अपनी वस्तु का मूल्य CP से 20% अधिक अंकित करता है। वह आधी वस्तु को अंकित मूल्य पर बेचता है। और $\frac{1}{4}$ भाग को 20% छूट पर तथा शेष ($\frac{1}{4}$) को 40% की छूट पर बेचता है। उसका लाभ % क्या होगा?

Solution :— Let CP = 100₹

MP = 120₹



$SP_1 = 60$ [No discount]

$SP_2 = 30 \times \frac{80}{100} = 24$ [20% discount]

$SP_3 = 30 \times \frac{80}{100} = 18$ [40% discount]

Total SP = 60 + 24 + 18 = 102 ₹

P% = 2% ANS.

FALSE WEIGHT / DISHONEST SHOPKEEPER
[बेईमान दुकानदार]

$$\text{PROFIT/LOSS \%} = \frac{\text{ERROR}}{\text{FALSE WT}} \times 100$$

where Error = True wt − False wt.

IF ANS is + ⇒ profit otherwise (−) LOSS

A dishonest shopkeeper promises to sell his goods at its CP. But he uses 800 gm wt. instead of 1 kg. Find Profit %?

Solution:— I (Basic)

Let CP of 1 gm = 1 ₹

CP of 800 gm = 800

but SP of 800 gm = 1000

$$\Rightarrow P\% = \frac{200}{800} \times 100 = 25\%$$

II (TRICKY)

$$\underset{800}{CP} \longleftarrow \boxed{1000} \longrightarrow \underset{1000}{SP}$$

$$\frac{200}{800} \times 100 = 50\%$$

एक दुकानदार अपनी वस्तुओं को क्रय मूल्य पर ही बेचने का वायदा करता है। लेकिन वह सामान देते वक्त 1 kg के भार के स्थान पर 1100 gm का भार गलती से इस्तेमाल कर लेता है। तो हानि% बताओ?

Solution:— I

$$\underset{1100}{CP} \longleftarrow \boxed{1000} \longrightarrow \underset{1000}{SP}$$

$$\text{LOSS \%} = \frac{100}{1100} \times 100 = 9.09\%$$

II

By formula

$$\text{Loss\%} = \frac{-100}{1100} \times 100$$

$$= 9.09\%$$

A shopkeeper promises to sell at CP. But by means of false weight he gains 25%. Find the weight he uses in place of 1kg?

Solution :— I (BASIC)

Let he cheats by x gm

$\Rightarrow \dfrac{x}{1000-x} * 100 = 25$

$\Rightarrow 4x = 1000 - x \Rightarrow x = 200$

True wt = 1000 − 200 = 800 Ans.

II (TRICKY)

$25\% = \dfrac{1 - P}{4 - CP} \Rightarrow SP = 5$

if 5 ⟶ 1000 gm

∴ 4 ⟶ 800 gm Ans

A shopkeeper promises to sells his goods at 10% profit but he uses 20% less weight. Find the profit %?

Solution :— I

$800 \xleftarrow[-20\%]{CP} \boxed{1000} \xrightarrow[10\%]{SP} 1100$

$\dfrac{300}{800} \times 100 = 37.5\%$

II $P\% = \dfrac{1 - P}{10 - CP} \Rightarrow SP = 11$

CP		SP	
10	:	11	भी
4	:	5	→ $20\% = \frac{1}{5}$ ← कुल वजन
40	:	55	

$15/40 \times 100 = 37.5\%$

एक दुकानदार अपने ग्राहक से वादा करता है कि प्रत्येक वस्तु को 44% की हानि पर बेचेगा लेकिन वह भार में 30% कम कर देता है। तो उसकी वास्तविक हानि % बताओ?

Solution :- I

$700 \xleftarrow{-30\%} \boxed{1000} \xrightarrow{-44\%} 560$

$\dfrac{140}{700} \times 100 = 20\%$ Ans.

II

	CP	:	SP
44% L →	100	:	56
30% भार कम →	70	:	100
	70	:	56

$\dfrac{14}{70} \times 100 = 20\%$

एक दुकानदार अपने ग्राहक से वादा करता है कि वह प्रत्येक वस्तु को x% की हानि पर बेचेगा लेकिन वह भार में 20% कम भार का प्रयोग करता है। तथा 20% का लाभ कमाता है तो x का मान ज्ञात कीजिए

Solution:- I

20% कम भार: CP : SP = 4 : 5

Loss% = x → a : b

कुल लाभ 4a → 5 : 6 ← 5b

$\frac{4a}{5b} = \frac{5}{6} \Rightarrow \frac{a}{b} = \frac{25}{24}$, $\frac{1}{25} \times 100 = 4\%$

II

CP $\boxed{800} \leftarrow \boxed{1000} \rightarrow 960$ SP (20% P)

$x\% = \frac{40}{1000} \times 100 = 4\%$ ANS.

एक दुकानदार सामान खरीदते वक्त 10% बेईमानी तथा सामान बेचते वक्त भी 10% की बेईमानी करता है। तो कुल लाभ % ज्ञात करो?

Solution:- I

$x + y + \frac{xy}{100}$

$10 + 10 + \frac{100}{100} = 21\%$ लाभ

OR

II

$1000 \xrightarrow{10\%} 1100 \xrightarrow{10\%} 1210$

$\frac{210}{1000} \times 100 = 21\%$ लाभ

#

A shopkeeper sells an article for ₹ 96 such that P% and CP are equal. Find CP?

Solution:- I (BASIC)

Let CP = P% = x

$\Rightarrow x \times \frac{(100+x)}{100} = 96$

$\Rightarrow x^2 + 100x = 9600$

$\Rightarrow x^2 + 100x - 9600 = 0$

$(x+160)(x-60) = 0$

$x = -160$ × or $\boxed{x = 60}$

II (TRICKY)

SP = 96 — अन्तर हमेशा 10 होना चाहिए

$96 \rightarrow 16,\ 6$

$16 - 6 = 10$

$6 \xrightarrow{\times 10} \boxed{60}$ ANS.

$\left[\begin{array}{l} a \times b = 96 \\ a - b = 10 \end{array}\right]$ then add 0 to smaller No. to get ANS.

III) Try to do these Q. by options

CP = 60, P% = 60 $\Rightarrow$ SP = 96

एक दुकानदार एक वस्तु को 16 ₹ में इस प्रकार से बेचता है कि हानि% और क्र०मू० बराबर हो तो उस वस्तु का क्रय मूल्य बताइए? a) 60 b) 70 ✓c) 80 d) 90

Solution:- I (BASIC)

Let CP = L% = x

$\Rightarrow x \times \frac{(100-x)}{100} = 16$

$\Rightarrow 100x - x^2 = 1600$

$\Rightarrow x^2 - 100x + 1600 = 0$

$\Rightarrow (x-80)(x-20) = 0$

$\boxed{x = 80, 20}$ both are ✓

II (TRICKY)

SP = 16

$a \times b = 16$

$a + b = 10$

$16 \rightarrow 8,\ 2$

$8 \xrightarrow{\times 10} \boxed{80}$ $\quad$ $2 \xrightarrow{\times 10} \boxed{20}$ both are ✓

III) By options check (c) CP = 80, L% = 80%

$\Rightarrow$ SP = 16 Ans (c)

A man buys some Articles for ₹ 4800. He sells $\frac{2}{5}$th of the articles at a Profit of 25%. At what Price should he sell the Remaining articles so that he gains 19% on a whole?

Solution :- I

Let Total Articles = 5

कुल	25% पर	शेष
5	2	3

$\times 19 \left(\quad \left(\times 25 \right. \quad \right) \div 3$

95% − 50% = 45% = 15% Ans.

II (Alligation)

X%		25%
	19%	
3	:	2

$\Rightarrow \frac{25-19}{19-x} = \frac{3}{2}$

$\Rightarrow x = 15\%$ Ans.

IIIrd) $\frac{2}{5}(25\%) + \frac{3}{5}(x\%) = 19\%$

$\Rightarrow 50\% + 3x\% = 95\% \Rightarrow x = 15\%$

In what Ratio must water be mixed with milk to gain 20% on selling the mixture at cost Price?

Solution :- Alligation (I)

Let CP of 1 ltr of milk = 1 ₹

CP of 1 ltr of water = 0

SP of 1 ltr of mixture = 1

Now CP of 1 ltr of mix $= \frac{100}{120} \times 1 = \frac{5}{6}$

w		m
0		1
	$\frac{5}{6}$	
1	:	5

w : m = 1 : 5

II (TRICKY)

$20\% = \frac{1}{5} \begin{matrix} \rightarrow w \\ \rightarrow m \end{matrix}$

w : m = 1 : 5

[क्योंकि जो Profit होगा वो पानी के कारण ही होगा]

IF selling price of an articles gets doubled profit will get triples? Fin original P%?

Solution :- BASIC (I)

Let $CP = X$, $SP = Y$

Profit $= (Y - X)$

New $SP = 2Y$ New $P = 2Y - X$

$\therefore\ 2Y - X = 3(Y - X)$

$\Rightarrow Y = 2X$ OR $X : Y = 1 : 2$

$P\% = \frac{1}{1} \times 100 = 100\%$ ANS.

TRICKY (II)

CP = same = constant

	Old	:	New
SP	$1_{\times 2}$	:	$2_{\times 2}$
P	$1_{\times 1}$	:	$3_{\times 1}$

2

SP	2] CP = 1	4] CP = 1
P	1	3

CP : SP = 1 : 2

$\Rightarrow P\% = 100\%$

III) By options $P\% = 100\%$

let CP = 100 → $SP_1 = 200$ $SP_2 = 400 \Rightarrow P = 300$ Ans 100%.

एक दुकानदार ने एक वस्तु को 20% के लाभ पर B को बेचा, B ने उसको 25% के लाभ पर C को बेच दिया। अगर C का CP 225 है तो A का CP ज्ञात करो?

Solution :- Note: जो B का CP है वो A का SP है।

I

$CP(A : B) = 100 : 120$
$= 5 : 6$

$CP(B : C) = 100 : 125$
$= 4 : 5$

$CP(A : B : C) = 20 : 24 : 30$

$30 \longrightarrow 225$

$\Rightarrow 20 \longrightarrow \frac{225}{30} \times 20 = 150$ ₹ Ans.

II

$CP(A) = 225 \times \frac{100}{120} \times \frac{100}{125}$
$= 150$ ₹ ANS.

III

$CP(A) = 225 \times \frac{5}{6} \times \frac{4}{5}$
$= 150$ ₹ Ans.

Selling Price Same

A) IF $SP_1 = SP_2$ $\Rightarrow$ Always $\Rightarrow$ $\boxed{\text{Loss\%} = \dfrac{x^2}{100}}$

$P\% = L\% = x\%$ Loss

B) IF $CP_1 = CP_2$ $\Rightarrow$ No Profit, No Loss

$P\% = L\%$

Two Books are sold for ₹ 960 each. One is sold at 20% Profit and other at 20% Loss. Find Profit and loss in ₹?

Solution :– I (BASIC)

$SP_1 + SP_2 = 960 + 960 = 1920$

$CP_1 = 960 \times \dfrac{100}{120} = 800$ ₹

$CP_2 = 960 \times \dfrac{100}{80} = 1200$ ₹

$CP_1 + CP_2 = 2000 \Rightarrow \text{Loss} = 80$ ₹

II (TRICKY)

$P\% = L\% = 20\%$

$SP_1 = SP_2 = 960$

$\Rightarrow \text{Loss\%} = \dfrac{(20)^2}{100} = 4\%$

$\text{Loss} = 1920 \times \dfrac{4}{96}$

$= 80$ ₹

राम के पास दो घोड़े हैं। जिनमें से वह प्रत्येक को समान विक्रय मूल्य ₹75000 में बेचता है। अगर वह एक को 20% लाभ पर और दूसरे को 10% की हानि पर बेचता है तो दोनों घोड़ों के क्रय मूल्यों के अनुपात ज्ञात करो?

Solution :– Trick

$$\frac{CP_1}{CP_2} = \frac{100 \pm x_2}{100 \pm x_1} \Rightarrow \frac{CP_1}{CP_2} = \frac{100-10}{100+20} = \frac{3}{4}$$

$CP_1 : CP_2 = 3 : 4$

Two horses are sold for ₹ 2.75 EACH IF one is sold at 20% Profit and other at 10% loss. Find the OVERALL PROFIT and loss % ?

Solution :- I 20% P ⟶ 120% of CP = SP

10% L ⟶ 90% of CP = SP

let $SP_1 = SP_2 = 36$ (LCM of 12 & 9)

$CP_1 = \frac{36}{1.20} = 30$, $CP_2 = \frac{36}{.90} = 40$

$CP_1 + CP_2 = 70$

$SP_1 + SP_2 = 72$

$P\% = \frac{2}{70} \times 100$

$= 2.85\%$ **ANS.**

II $20\% P = \frac{1 \to P}{5 \to CP}$ $SP_1 = 6 \times 3 = 18$

$10\% L = \frac{1 \to L}{10 \to CP}$ $SP_2 = 9 \times 2 = 18$

$CP_1 \to 5 \times 3 = 15$
$CP_2 \to 10 \times 2 = 20$] 35 $SP = 36$

$P\% = \frac{1}{35} \times 100 = 2.85\%$ **ANS.**

एक आदमी दो वस्तुओं को 1710 ₹ में बेचता है अगर पहली वस्तु का क्रय मूल्य दूसरी वस्तु के विक्रय मूल्य के बराबर है तो और पहली को 10% की हानि पर और दूसरी को 25% के लाभ पर बेचा जाए तो उसे कुल कितने ₹ की लाभ या हानि होगी?

Solution:— I

	1st	2nd	
CP	$\boxed{100}$	80	= 180
	↓ –10% L	↓ 25% P	10₹ P
SP	90	$\boxed{100}$	= 190
	190	180	

$SP_1 + SP_2 \to 190 \longrightarrow 1710$

$1 \longrightarrow 9$

$10 \longrightarrow 90₹$ Ans

II (Ratio)

10% L CP_1 SP_1

10 : 9

25% P CP_2 : SP_2

4 : 5

but $CP_1 = SP_2 = 10$

10 : 9

$4_{\times 2}$: $5_{\times 2}$

$SP_1 + SP_2 = 10 + 9 = 19$

$19 — 1710 \Rightarrow 1 \to 90₹$ Ans.

A man sells two articles first at 15% loss and 2nd at 19% profit. If during the whole transaction he bears a loss of 90₹. If he sells both article at same price Find the cost price of each?

Solution :– 15% L $\Rightarrow \frac{-3}{20}$

$CP_1 : SP_1$

20 : 17

19% P $\rightarrow \frac{19}{100}$

$CP_2 : SP_2$

100 : 119

but $SP_1 = SP_2$

	I	II	
CP	$20_{\times 7}$	100	= 240
P/L	$3_{\times 7}$	19	2
SP	$17_{\times 7}$	119	= 238

$2 \longrightarrow 90$

$1 \longrightarrow 45$

$CP_1\,(140) \to 45 \times 140$ 6300

$CP_2\,(100) \to 45 \times 100$ 4500

A man purchased two watches for ₹ 840. If he sells one at 16% profit and other at 12% loss then he get no profit no loss in whole Transaction. Find the CP of each watch?

Solution :- I (BASIC)

Let $CP_1 = x \Rightarrow CP_2 = 840 - x$.

$x \times \frac{88}{100} + (840-x) \times \frac{116}{100} = 840$

$x = 480 = CP_1 \Rightarrow CP_2 = 360$

II (Alligation)

$-12 \qquad 16$

0

$16 : 12$

$4 : 3$

$CP_1 = \frac{840}{7} \times 4 = 480$

$CP_2 = 360$

III (TRICKY)

$16\,CP_1 = 12\,CP_2$

$\Rightarrow \frac{CP_1}{CP_2} = \frac{3}{4}$

$7 \longrightarrow 840$

$1 \longrightarrow 120$

$CP_1(3) - 360$

$CP_2(4) - 480$

एक व्यक्ति दो वस्तुओं को 1250 ₹ में खरीदता है। वह एक को 5% की हानि पर और दूसरे को 20% के लाभ पर बेचता है। अगर उसे पूरे व्यवहार क्रम में न लाभ और न ही हानि होती है। तो दोनों वस्तुओं का अलग-2 क्रय मूल्य ज्ञात करो?

Solution I (BASIC)

let $CP_1\,(5\%\,L) = x$

$\Rightarrow CP_2\,(20\%\,P) = 1250 - x$

$x \times \frac{95}{100} + (1250-x)\frac{120}{100} = 1250$

$\Rightarrow \frac{-25x}{100} = -1250 \times \frac{20}{100}$

$x = 1000$

$CP_2 = 1250 - 1000 = \underline{250}$

II (Alligation)

$-5\% \qquad 20\%$

0

$20 : 5$

$4 : 1$

$5 \longrightarrow 1250$

$4 \longrightarrow 1000\,(CP_1)$

$1 \longrightarrow 250\,(CP_2)$

III (TRICKY)

$5\,CP_1 = 20\,CP_2$

$\frac{CP_1}{CP_2} = \frac{4}{1}$

$5 \longrightarrow 1250$

$4 \longrightarrow 1000\ CP_1$

$1 \longrightarrow 250\,(CP_2)$

A shopkeeper sells table at 25% profit and chair at 20% loss then he gains 18₹. But if he sells table at 20% loss and chair at 25% P then there is no profit no loss situation. Find the cost price of chair and table?

Solution:— No Profit No loss

$20T = 25C$

$T : C = 5 : 4$

$CP \longrightarrow 500 : 400$

$400 \times \frac{20}{100} = 80$ ₹ loss

$500 \times \frac{25}{100} = 125$ Profit

40 Profit ↑

$45 \longrightarrow 18$

$1 \longrightarrow \frac{2}{5}$

CP of Table

$C \rightarrow 500 \times \frac{2}{5} = 200$

CP of chair

$400 \rightarrow 400 \times \frac{1}{5}$

$= 100$ ₹

एक पेन को 5% की हानि पर और एक किताब को 15% के लाभ पर बेचने पर एक दुकानदार को 7 ₹ का फायदा होता है। यदि वह पेन को 5% लाभ पर और किताब को 10% लाभ पर बेचता तो उसे 13 ₹ का लाभ होता। किताब का वास्तविक मूल्य ज्ञात करो?

Solution

पेन	किताब	
~~5% (L)~~	15% P	= 7 ₹ P
~~5% (P)~~	10% P	= 13 ₹ P

$25\% \rightarrow 20$

$4\times (\quad) \times 4$

$100\% \quad 80$ Ans.

Total cost of 8 books and 5 pens is 92. And cost of 5 books and 8 pens is 77. then find the cost of 3 books and 2 pens?

Solution:-

$8B + 5P = 92$ — (I)

$5B + 8P = 77$ — (II)

Add (I)+(II), we get

$13B + 13P = 169 \Rightarrow B+P = 13$ — (3)

Subtract (I) and (II)

$3B - 3P = 15$

$B - P = 5$ — (4)

adding 3 and (4) $B = 9,\ P = 4 \Rightarrow 3B + 2P = 3\times9 + 2\times4 = 35$

The selling price of X and Y are ₹ ₹100 each. X calculates his profit on SP while Y on CP. Find the difference between their CP if both claims 20%.

Solution:-

	X	Y
CP	$4_{\times6}$	$5_{\times5}$
P/L	$1_{\times6}$	$1_{\times5}$
SP	$5_{\times6}$	$6_{\times5}$

$\Rightarrow D = 1$

$30 \longrightarrow ₹100$

$1 \longrightarrow$ 70 Ans.

A shopkeeper bought two bicycles in ₹ 1600. If he sold 1st bicycle at 10% profit and 2nd at 20% profit he earns a certain profit. but if he sold 1st at 20% P and 2nd at 10% P he will get ₹ 5 more. Find the CP of bicycles?

Solution:-

$B_1 + B_2 = 1600$

$B_1 - B_2 = 50$

$$\left[\begin{array}{l} 10\%\cdot B_1 + 20\%\cdot B_2 = P \\ 20\%\cdot B_1 + 10\%\ B_2 = P+5 \\ \hline B_1 - B_2 = \frac{5\times100}{10} = 50 \end{array}\right]$$

$2B_1 = 1650$

$\Rightarrow B_1 = 825 \Rightarrow B_2 = 775$

CP of 12 oranges is equal to SP of 9 oranges and discount on 10 oranges is equal to profit on 5 oranges. What is the percent point difference between the profit % and discount %?

a) $16\frac{2}{3}\%$ b) 25 % c) 22.22 % d) 20 %

Solution :- $12\,CP = 9\,SP$

$\Rightarrow \frac{CP}{SP} = \frac{3}{4} \Rightarrow P\% = \frac{1}{3} \times 100 = 33.33\%$

Now $10D = 5P$

$\frac{D}{P} = \frac{1}{2} \Rightarrow$ If $P \to 1$, $D \to \frac{1}{2}$

CP	SP	MP
3	4	4.5

$3 \xrightarrow{+1} 4 \xrightarrow{+1/2} 4.5$

$D\% = \frac{4.5 - 4}{4.5} \times 100 = 11\frac{1}{9}\%$

$P\% - D\% = \frac{100}{3} - \frac{100}{9} = \frac{200}{9} = 22.22\%$ Ans.

The ratio of cost price and marked price of an article is 2:3 and the ratio of percentage profit and percentage discount is 3:2. What is D%?

a) 16.66 % b) 20 % c) 25 % d) 33.33 %

Solution: Let $P\% = 3x$ and $D\% = 2x$ [$\because$ 3:2]

CP : MP

$100 - D\%$: $100 + P\%$

$100 - 2x$: $100 + 3x$

$\frac{100 - 2x}{100 + 3x} = \frac{2}{3} \Rightarrow 300 - 6x = 200 + 6x$

$\Rightarrow 12x = 100 \Rightarrow x = \frac{25}{3}$

$D\% = 2x = 2 \times \frac{25}{3} = \frac{50}{3} = 16.66\%$

10 मोमबत्तियों को बेचने पर एक व्यक्ति को 3 कलम के विक्रय मूल्य के बराबर लाभ होता है। जबकि 10 कलम को बेचने पर 4 मोमबत्तियों के विक्रय मूल्य के बराबर हानि होती है। यदि लाभ % और हानि % संख्यात्मक बराबर हो और मोमबत्तियों का क्रय मूल्य कलम के क्रय मूल्य का आधा हो तो मोमबत्ती तथा कलम के विक्रय मूल्य का अनुपात ज्ञात करो?

Solution :- मोमबत्ती कलम

CP 1 : 2

SP a : b

$CP = 10$
$P = 3b$
$P\% = \frac{3b}{10} \times 100$

$CP = 10 \times 2$
$= 20$
$L = 4a$
$L\% = \frac{4a}{20} \times 100$

$$\frac{3b \times 100}{10} = \frac{4a}{20} \times 100$$

$$\frac{3b}{10} = \frac{4a}{20}$$

$a : b = 3 : 2$ Ans

एक व्यक्ति तीन वस्तुओं को समान विक्रय मूल्य पर बेचता है। वह पहली वस्तु को 20% के लाभ पर, दूसरी 10% की हानि पर तथा तीसरी को 25% की हानि पर बेचता है। इस प्रकार पूरे सौदे में उसे कुल मिलाकर 5 रु की हानि होती है। तो क्रय मूल्य ज्ञात करो?

Solution :- 20% ↑ 10% ↓ 25% ↓

$+\frac{1}{5}$ $-\frac{1}{10}$ $-\frac{1}{4}$

$SP_1 = 6$ $SP_2 = 9$ $SP_3 = 3$

but $SP_1 = SP_2 = SP_3 = 18$ (LCM)

CP SP

$15 = 5 \times 3$ $6 \times 3 = 18$

$20 = 10 \times 2$ $9 \times 2 = 18$

$24 = 4 \times 6$ $3 \times 6 = 18$

$59 \xrightarrow{-5} 54$

$5 \longrightarrow 120$

$1 \longrightarrow 24$

$CP(59) \xrightarrow{\times 24} 1416$

$CP_1 \longrightarrow 15 \times 24 = 360$

$CP_2 \longrightarrow 20 \times 24 = 480$

\# A sells an article to B at a profit of 10%. and B sells the article back to A at a loss of 10%. find the over all profit and loss % of A.

a] A neither gains nor losses b] A loss 1 %

c] A makes a profit of 20% d] A makes a profit of 11%

Solution :– I

let CP of A = 100

CP for B = 100+10 = 110

Now B sells it at ~~gain~~ loss of 10% to A

Now CP of A this time = 99

Total Profit of A = 11

P%

II

Simple Interest

SIMPLE INTEREST
(साधारण ब्याज)

SI/CI are Extension of Percentages.
"Credit the giver, Debit the Receiver"

SI (साधारण ब्याज) $= \frac{P \times R \times T}{100}$

where P → Principle (मूलधन)
R → Rate of Interest (दर)
T → Time (समय)

Amount (मिश्रधन)

$$\boxed{A = P + SI/CI}$$

Rate (R%) :→ Interest for 1 year for every 100 ₹ is called Rate of interest.
5% Rate means → ₹5 interest on 100 for 1 yr

In simple Interest Principle (मूलधन) हमेशा same रहता है। OR
Simple Interest remains same for same period of time. e.g. If SI of 2yrs = 20
⇒ SI of 4 yrs = 40
SI of 8 yrs = 80

Note :— whenever it is not mentioned whether we have to take CI OR SI we should take SI.

\# Principle is always 100%

if Rate of interest is R%

$\Rightarrow$ interest for 1 yr is R% of P

" " 2 yrs is 2R% of P

" " 10 yrs is 10R% of P

" " T yrs " TR% of P

generally SI = RT% of P

$\hookrightarrow$ Effective Rate

\# e.g.

IF R = 5%
T = 6 years

$\rightarrow$ P = 100%

$\rightarrow$ SI = 30%

$\rightarrow$ A = 130%

\# IF P = 15625, R = 2.5% p.a., T = 4 yrs, SI = ?

Solution :$\rightarrow$ effective Rate = RT% = 2.5 × 4 = 10%

P(मूलधन) — 100% $\longrightarrow$ 15625

SI — RT% — 10% $\longrightarrow$ 1562.5 ₹ Ans.

\# R = 4%, T = 5 yrs, SI = 80, P = ?

Solution :- effective Rate = RT% = 4×5 = 20%

SI $\rightarrow$ 20% $\longrightarrow$ 80

P $\rightarrow$ 100% $\longrightarrow$ $\frac{80}{20} \times 100$ = 400 ₹ Ans.

Amount (मिश्रधन) $= P + SI$
$$= 100\% + RT\%$$
$$= (100 + RT)\%$$

$R = 5\%$, $T = 4$ yrs, $P = 2000$, $A = ?$

Solution :– $P = 100\%$

$RT\% = 5 \times 4 = 20\%$ $\Rightarrow A = 120\%$

(P) $100\% \longrightarrow 2000$

(A) $120\% \longrightarrow \frac{2000}{100} \times 120 = 2400$ ₹ ANS.

$R = 5\%$, $T = 4$ yrs, $A = 480$, $P = ?$

Solution :– $P = 100\%$ $RT\% = 5 \times 4 = 20\%$

$\Rightarrow A = 120\%$

A $(120\%) \longrightarrow 480$

P $(100\%) \longrightarrow \frac{480}{120} \times 100 = 400$ ₹ ANS.

IN how many years ₹ 500 will Amount to ₹ 625 at 5% P.a.

Solution :– $A = \underset{\substack{\downarrow \\ 100\%}}{P} + SI$

P $(500) \longrightarrow 100\%$

A $(625) \longrightarrow \frac{100\%}{500} \times 625 = 125\%$

$\Rightarrow SI = 125\% - 100\% = 25\% = RT\%$

but $R = 5\%$ $\Rightarrow T = \frac{25}{5} = 5$ yrs ANS.

एक निश्चित राशि 4 साल में 360 और 10 साल में 450 हो जाती है। तो मूलधन और ब्याज की दर बताओ?

Solution:-

$$\begin{array}{ll} 10 \text{ years} & = 450 \\ 4 \text{ year} & = 360 \\ \hline 6 \text{ year} & = 90 \end{array} \Rightarrow 1 \text{ year} = 15$$

Principle $= 360 - 4\times 15 = 300$

$$\frac{300\times R\times 1}{100} = 15 \Rightarrow R = 5\%$$

एक निश्चित राशि 2 years में 720 रु और 7 साल में 1020 रु हो जाती है तो मूलधन और ब्याज की दर ज्ञात करें?

Solution:-

7 साल में ब्याज = 1020

2 साल में ब्याज = 720

5 साल में ब्याज = 300

1 साल में ब्याज $\frac{300}{5} = 60$

Principal $= 720 - 60\times 2 = 600$

$$\frac{600\times 1\times R}{100} = 60 \Rightarrow R = 10\%$$

कोई धन राशि साधारण ब्याज की दर से 4 साल के लिए कर्ज पे दी जाती है। यदि ब्याज की दर 5% कम होता तो 400 रु कम ब्याज मिलता तो मूलधन ज्ञात करो?

Solution:-

20% —— 400

1% —— 20

100% —— 2000 Ans.

₹ 425 amounts to ₹ 476 in 3 yrs at certain Rate of interest. Find Rate of interest?

Solution :— $A = \underset{\substack{\downarrow \\ 100\%}}{P} + SI$

P $\quad 425 \longrightarrow 100\%$

A $\quad 476 \longrightarrow \dfrac{100\%}{425} \times 476 \longrightarrow 112\%$

$SI = A - P = 112\% - 100\% = 12\% = RT\%$

but $T = 3$ yrs $\Rightarrow R\% = \dfrac{12}{3} = 4\%$ ANS.

A certain principle in four years will become 3 times. Find the Rate?

Solution :— I (FORMULA)

$P = x \quad A = 3x$

$\underbrace{\quad}_{SI}$ $2x$

$2x = \dfrac{x \times 4 \times R}{100}$

$\Rightarrow R = 50\%$ ANS

II (CONCEPT)

$A = P + SI$

$P = 100\% \quad A = 300\%$

$\Rightarrow SI = 200\% = RT$

but $T = 4$ yrs

$\therefore R = \dfrac{200}{4} = 50\%$

III (DIRECT)

$R = \dfrac{(N-1)}{T} \times 100$

$= \dfrac{3-1}{4} \times 100$

$= 50\%$

कोई राशि 5% वार्षिक साधारण ब्याज की दर से एक निश्चित समय में दुगुनी हो जाती है। समय ज्ञात कीजिए?

Solution :— I (FORMULA)

$P = x \quad A = 2x$

$\underbrace{\quad}_{x = SI}$

$x = \dfrac{x \times 5 \times T}{100}$

$\Rightarrow T = 20$ yrs ANS

II (CONCEPTUAL)

$A = P + SI$

$P = 100\% \quad A = 200\%$

$SI = 100\% = RT\%$

but $R = 5\%$

$\Rightarrow T = \dfrac{100}{5} = 20$ yrs

III (DIRECT)

$R = \dfrac{N-1}{T} \times 100$

$5 = \dfrac{2-1}{T} \times 100$

$\Rightarrow T = 20$ yrs

अगर समय महीनों में या दिनों में दिया गया हो तो उसको वर्ष में बदलना होगा।

e.g. 6 month $= \frac{6}{12} = \frac{1}{2}$ वर्ष   3 month $= \frac{3}{12} = \frac{1}{4}$ वर्ष

73 days $= \frac{73}{365} = \frac{1}{5}$ वर्ष, 146 days $= \frac{146}{365} = \frac{2}{5}$ वर्ष

219 days $= \frac{219}{365} = \frac{3}{5}$ वर्ष, 292 days $= \frac{292}{365} = \frac{4}{5}$ वर्ष

6000 रु० पर 6% वार्षिक दर से 8 माह का साधारण ब्याज तथा मिश्रधन ज्ञात करो?

Solution :- I (FORMULA)

$$SI = \frac{P \times R \times T}{100}$$

$$= \frac{6000 \times 6 \times 8}{100 \times 12} = 240$$

$$A\text{(मिश्रधन)} = \overset{P\ +\ SI}{6000 + 240}$$

$$= 6240/-$$

II (CONCEPTUAL)

$T = \frac{8}{12} = \frac{2}{3}$ year, $R = 6\%$

$RT\% = \frac{2}{3} \times 6 = 4\%$

$SI \rightarrow 4\%$ of 6000

$= 240$

$A = 6000 + 240$

$= 6240$

3200 रु० पर 4 अप्रैल 2012 से 16 जून 2012 तक का 5% वार्षिक दर पर साधारण ब्याज कितना होगा?

Solution :→ Days $= \overset{April}{26} + \overset{may}{31} + \overset{June}{16}$

$= 73$ दिन

Time $= \frac{73}{365} = \frac{1}{5}$ वर्ष

Note :- पहले दिन का ब्याज नहीं लगता लेकिन अंतिम दिन पर लगता है

$$SI = \frac{3200 \times 5}{100} \times \frac{1}{5} = 32 \text{ रु०}$$ ANS.

A sum of money doubles in 8 yrs at the rate of simple interest. In how much time it will become 32 times?

Solution:— I (SIMPLE)

P [x] —— x (SI) —— A [2x]

↓ 8 yrs

P [x] —— 31x (SI) —— A [32x]

↓

31×8 = 248 yrs ANS.

II (TRICKY)

$$\frac{(N_1-1)}{(N_2-1)} = \frac{T_1}{T_2}$$

$$\frac{2-1}{32-1} = \frac{8}{T_2} \Rightarrow T_2 = 248$$

कोई राशी 4 साल में 3 गुणी हो जाती है। वही राशी साधारण ब्याज की दर से 7 गुणी होने में कितना समय लेगी ?

Solution:— I (Simple)

माना मूलधन = 1 ₹

मिश्रधन = 3 ₹

SI = 3 − 1 = 2 ₹

2 ₹ SI ⟶ 4 साल

7 गुणा में 1 अपना है।

SI → 7 − 1 → 6 ₹

2 — 4 साल

6 — $\frac{4}{2}\times 6$ = 12 साल उत्तर

II (TRICKY)

$$\frac{(N_1-1)}{(N_2-1)} = \frac{T_1}{T_2}$$

$$\frac{3-1}{7-1} = \frac{4}{T_2}.$$

$$\Rightarrow T_2 = 12 \text{ yrs}$$

कोई राशी 5 साल में साधारण ब्याज की दर से 5 गुणी हो जाती है। तो वही राशी कितने समय में 7 गुणा हो जायगी ?

Solution:— $\frac{(N_1-1)}{(N_2-1)} = \frac{T_1}{T_2} \Rightarrow \frac{5-1}{7-1} = \frac{5}{T_2}$

$\Rightarrow T_2 = 7.5$ साल

\# A sum of money was invested at SI at a certain rate for 3 years. Had it been invested at 4% higher rate. It would have fetched ₹ 480 more. Find the principal?

Solution :- Additional money obtained

$= 4\%$ for 3 yrs $= 12\%$ of P

$12\% \longrightarrow 480$

$100\%\ (P) \longrightarrow \frac{480}{12} \times 100 = 4000$ ₹ **ANS.**

\# A certain sum of money amounts to ₹ 704 in 2 yrs and ₹ 800 in 5 yrs. Find the Principal and rate?

Solution :-

A_5		A_2
$\boxed{800}$	3 YRS SI	$\boxed{704}$
	96 ₹	

interest for 3 yr $= 96$

" " 1 yr $= \frac{96}{3} = 32$

" " 2 yrs $= 32 \times 2 = 64$

$\therefore P = A_2 - SI_2 = 704 - 64 = 640$ **ANS.**

Rate (R) $\% = \frac{32 \times 100}{640 \times 1} = 5\%$ **ANS**

If the Rate of interest for first 3 years is 6% p.a. for the next 4 years is 7% p.a. and for the period beyond 7 years is 7.5% p.a. If a man lent out ₹ 1200 for 11 years. Find the total Interest earned by him?

Solution :- 3 yrs — $6\% \times 3 = 18\%$

4 yrs — $7\% \times 4 = 28\%$

4 yrs — $7.5\% \times 4 = 30\%$

76% of P

Simple Interest $= 76\%$ of $P = 76\%$ of 1200

$= (75\% + 1\%)$ of 1200

$= \frac{3}{4} \times 1200 + 12 = 912$ Ans.

एक व्यक्ति एक निश्चित धनराशि बैंक में जमा करता है। यदि ब्याज की दर प्रथम दो वर्ष 3% वार्षिक और अगले तीन वर्ष तक 8% वार्षिक और अन्तिम 1 वर्ष 10% वार्षिक साधारण ब्याज हो तो और अगर 6 साल में कुल 1520 रू. साधारण ब्याज प्राप्त हुआ हो तो मूलधन ज्ञात करो?

Solution :- 2 yrs $\longrightarrow 2 \times 3\% = 6\%$

3 yrs $\longrightarrow 3 \times 8\% = 24\%$

1 yrs $\longrightarrow 1 \times 10\% = 10\%$

SI $\longrightarrow 40\%$ of P

$40\% \longrightarrow 1520$

$100\% (P) \longrightarrow \frac{1520}{40} \times 100 = 3800$ Ans.

\# A man lent 2000 partially at 5% p.a. and balance at 4% p.a. If he receives ₹ 92 as annual interest. Find the amount lent at 5%?

Solution:– I (BASIC)

Let x ₹ invested @ 5%

$\Rightarrow$ 5% of x + (2000 − x) 4% = 92

$\Rightarrow$ 5x − 4x − 8000 = 9200

$\Rightarrow$ x = 1200 Ans

II (Alligation)

$R\% = \frac{92}{2000} \times 100 = 4.6\%$

4% 5%

4.6%

.4 .6

2 : 3

$= \frac{2000}{5} \times 3$

= 1200

III (रामबाण)

Let whole amnt @ 4%

$SI = \frac{2000 \times 4}{100} = 80$

gap = 92 − 80 = 12

5% − 4% $\longrightarrow$ 12

1% $\longrightarrow$ 12

100% $\longrightarrow$ 1200

\# A man lent ₹ 16000 part of which he lent at 4% and rest at 5% p.a. SI. If the total interest received was ₹ 700 in one year, the money lent at 4% was?

Solution:– I (Alligation)

overall R% $= \frac{700}{16000} \times 100$

$= \frac{350}{80}$

80 × 4% 5% × 80

$\frac{350}{80}$ × 80

50 30

5 : 3

at 4%

$= \frac{16000}{8} \times 5$

= 10,000 Ans.

II (रामबाण)

let all (16000) @ 5%

$SI = \frac{16000 \times 5}{100} = 800$

gap 800 − 700 = 100

5% − 4% $\longrightarrow$ 100

1% $\longrightarrow$ 100

100% $\longrightarrow$ 10000

(जिसको मानेंगे वो नहीं आयगा दूसरा आयगा)

\# Sunil borrowed a sum of 30,000. He took a part of it at 12% and remaining at 10%. At the end of two years he returned 36480. what amount was borrowed at 12% Rate?

Solution:– SI for 2 yr = 6480

" " 1 yr = 3240

let all (30000) invested at 10%

SI = 30,000 × 10% = 3000

gap = 240

12% − 10% = 2%

2% $\longrightarrow$ 240

100% $\longrightarrow$ 12000

(जिसको मानेंगे वो नहीं आयगा)

\# A sum of ₹ 14400 is divided into 3 parts such that 1st part is invested at 2% p.a for 3 years 2nd part @ 3% p.a. for 4 years and 3rd part at 4% p.a. for 5 years. If simple interest of all the 3 parts are equal find Principal of each part?

Solution:— let principal of 3 parts is P_1, P_2 & P_3

$$SI_1 \qquad SI_2 \qquad SI_3$$

$$\text{let } 6\% \text{ of } P_1 = 12\% \text{ of } P_2 = 20\% \text{ of } P_3 = 60 \text{ (माना)}$$

$$\Rightarrow P_1 : P_2 : P_3 = 10 : 5 : 3$$

$$P_1 = \frac{14400}{18} \times 10 = \underline{8000},\ P_2 = \frac{14400}{18} \times 5 = \underline{4000},\ P_3 = \underline{2400}$$

\# A sum of 18750 is left by will by a father to be divided between two sons of 12 and 14 years of age so that when they attain maturity at 18, the amount received by each at 5% p.a. SI will be same. Find the sum alloted to both at present.

Solution:— $A_1 \qquad A_2$

$$(100 + 5\% \times 6) \text{ of } P_1 = (100 + 5\% \times 4) \text{ of } P_2$$

$$130\% \text{ of } P_1 = 120\% \text{ of } P_2$$

$$13P_1 = 12P_2 \Rightarrow \frac{P_1}{P_2} = \frac{12}{13} \text{ or } P_1 : P_2 = 12 : 13$$

$$\therefore P_1 = \frac{18750}{25} \times 12 = \underline{9000},\ P_2 = \frac{18750}{25} \times 13 = \underline{9750}$$

with a given Rate of interest, the Ratio of the principal and Amount for a certain period of the time is 4:5. After 3 years with the same Rate of interest the ratio of Principal and amount becomes 5:7. The Rate of interest is ?

Solution:- [Principal is always same in SI]

P	A	I	Time
4	5	1	T yrs
5	7	2	T+3yr

we will make P same [LCM 20]

P	A	I	
20 = 4×5	5×5 = 25	5	T
20 = 5×4	7×4 = 28	8	T+3

I on ₹ 20 for 3yr = 8-5
= 3

$$R\% = \frac{3\times100}{20\times3} = \underline{\underline{5\%}}$$

if a Person invested some amount at the rate of 12% p.a. SI and remaining at 10% p.a. SI, he receives yearly interest ₹130. Had he interchanged the amount invested he would have received ₹134. How much money did he invest at different rates?

a). 500 @ 10%, 800 @ 12% b) 700 @ 10%, 600 @ 12%
c] 800 @ 10%, 400 @ 12% d) 700 @ 10%, 500 @ 12%

Solution:- I (BASIC)

12% of X + 10% of Y = 130
10% of X + 12% of Y = 134

⇒ X = 500, Y = 700

option D is correct

II (TRICKY)

Diff. in 2 Rates = 2%
Diff in Interest = ₹4
Diff in Amount $= \frac{4}{2}\times100$
= 200

only diff of 200 in option D

D is ✓

एक निश्चित धन राशि, एक निश्चित समय के लिए, एक निश्चित साधारण ब्याज की दर पर निवेश किया जाता है। यदि मूलधन को 20% बढ़ा दिया जाए और उसके दर को $\frac{6}{5}$ गुणा कर दिया जाए और समय को $2/3$ कर दिया जाए तो नया साधारण ब्याज ₹400 हो जाता है। प्रारंभिक साधारण ब्याज ज्ञात करें।

Solution:- [जो Effect Principal, Rate, Time पर पड़ेगा वही effect ब्याज पर पड़ेगा]

$$\therefore \text{New SI} = \text{old SI} \times \frac{120}{100} \times \frac{6}{5} \times \frac{2}{3}$$

$$₹400 = \text{old SI} \times \frac{120}{100} \times \frac{6}{5} \times \frac{2}{3}$$

$$\Rightarrow \text{old SI} = ₹500 \text{ ₹} \ \underline{\text{Ans.}}$$

एक निश्चित धनराशि एक निश्चित समय के लिए के लिए निवेश की जाती है। यदि साधारण ब्याज की दर 5% वार्षिक होता तो मिश्रधन 80 ₹ होता, जब ब्याज की दर 2% होता तो मिश्रधन 40 रु० होता तो समय क्या था?

Solution:- I (BASIC)

Let $P = 100\%$

$SI = RT\% = 5t\%$ & $2t\%$

$$\frac{100+5t}{100+2t} = \frac{80}{40}$$

$$\Rightarrow 100 + 5t = 200 + 4t$$

$$\Rightarrow \boxed{t = 100 \text{ yrs}}$$

II (TRICKY)

$$t = \frac{A_1 - A_2}{A_1R_2 - A_2R_1} \times 100$$

$$= \frac{80 - 40}{200 - 160} \times 100$$

$$= \frac{40}{40} \times 100$$

$$= \underline{100 \text{ yrs}}$$

INSTALLMENT [किश्त]

\# $$x = \frac{100D}{100T + \frac{R \times T[T-1]}{2}}$$

where $x \longrightarrow$ installment [equal]

$T \longrightarrow$ Time $\quad R \longrightarrow$ Rate (दर)

$D \rightarrow$ Debt (ऋण) Amount

\# OR Equate Future Value (Amount) मिश्रधन

FOR MONTHLY you can use it

$$P + \frac{P \times R \times T}{12 \times 100} = X * T + \frac{R * X}{12 \times 100}\left[1 + 2 + \text{——} + [T-1]\right]$$

\# what annual installment will discharge a debt of ₹ 2360 due in four years at 12% p.a. simple interest?

Solution:- I (FORMULA)

$$x = \frac{100D}{100T + \frac{RT(T-1)}{2}}$$

$$= \frac{100 \times 2360}{100 \times 4 + \frac{12 \times 4 \times 3}{2}}$$

$$= \frac{2360 \times 100}{472} = 500 \text{ ₹ } \underline{\text{ANS.}}$$

II Let each $I = 100$

[पहली किश्त पर 3 साल, दूसरी पर 2 साल, तीसरी पर 1 साल और चौथी पर 0 साल का ब्याज लगेगा]

$100 + 12 \times 3 = 136\%$

$100 + 12 \times 2 = 124\%$

$100 + 12 \times 1 = 112\%$

$100 + 12 \times 0 = 100\%$

$\underline{\quad 472\% \quad}$

$\therefore 472\% \longrightarrow 2360$

$100\% \longrightarrow 500$ $\underline{\text{ANS.}}$

1740 ₹ की राशी 5 बराबर वार्षिक किश्तों में देय है। यदि प्रत्येक किश्त पर शेष समय के लिए 8% की दर से साधारण ब्याज देय हो तो प्रत्येक किश्त की राशी ज्ञात करो?

Solution :— I (FORMULA)

$$x = \frac{100 \times 1740}{100 \times 5 + \frac{x 8 \times 5 \times 4}{2}}$$

$$= \frac{100 \times 1740}{580} = 300 ₹$$

II माना प्रत्येक किश्त = 100%

$100 + 4 \times 8\% = 132\%$
$100 + 3 \times 8\% = 124\%$
$100 + 2 \times 8\% = 116\%$
$100 + 1 \times 8\% = 108\%$
$100 + 0 \times 8\% = 100\%$
$\overline{\quad 580\% \quad}$

$580\% \longrightarrow 1740$

$\Rightarrow 100\% \longrightarrow \frac{1740}{580} \times 100 = 300$

IIIrd

$$5x + \frac{x \times 8}{100}[1+2+3+4] = 1740$$

$\Rightarrow 58x = 17400$

$\Rightarrow x = 300 ₹$

12800 की राशी 3 वार्षिक किश्तों में देय है। पहली किश्त की राशि दूसरी किश्त की राशि की आधी और तीसरी की 1/3 है। यदि प्रत्येक किश्त पर शेष समय के लिए 10% वार्षिक ब्याज देय हो तो प्रत्येक किश्त की राशि ज्ञात करो?

Solution :— $P_1 : P_2 : P_3$ we will find first

$$P_1 = \frac{P_2}{2} = \frac{P_3}{3} \Rightarrow P_1 : P_2 : P_3 = 1 : 2 : 3$$

OR 100 200 300

$100 + \frac{100 \times 10 \times 2}{100} = 120\%$

$200 + \frac{200 \times 10 \times 1}{100} = 220\%$

$300 + 0 = 300\%$

$\overline{\quad 640\% \quad}$

[पहली किश्त पर 2 वर्ष का, दूसरी पर 1 वर्ष का और तीसरी पर 0 वर्ष का ब्याज लगेगा]

$640\% \longrightarrow 12800$

$100\% \longrightarrow 2000$

$200\% \longrightarrow 4000$

$300\% \longrightarrow 6000$

Cash price of a mobile is 1500 ₹. But a man purchased for ₹ 350 cash down payment and 3 equal monthly installment of ₹ 400 each. Find the Rate of Interest?

Solution :— Rem^{I} = 1500 − 350 = 1150

Find equivalent 1 month P

1150 − 400
750 − 400
350
2250

also 400×3 = 1200
but Rem = 1150
⇒ SI = 50

$\therefore \frac{2250 \times R \times 1}{100 \times 12} = 50 \Rightarrow R = 26\frac{2}{3}\%$

OR

$P + \frac{PRT}{12 \times 100} = xT + \frac{Rx}{12 \times 100}[1+2+\ldots(T-1)]$

$1150 + \frac{1150 \times R \times 3}{12 \times 100} = 400 \times 3 + \frac{400R}{1200}\left[\frac{2 \times 3}{2}\right]$

$1150 + \frac{115R}{40} = 1200 + R$

$\Rightarrow 75R = 2000$

$R = 26\frac{2}{3}\%$ Ans.

1 व्यक्ति 10 रु० कर्ज लेता है और वह 1 रु० मासिक किश्तों में 11 महीनों में चुकाता है। ब्याज की दर क्या होगी?

Solution:— equivalent Principle for 1 month

पहले महीने 10 ₹ पर, दूसरे महीने 9 ₹ पर and so on ब्याज लगेगा क्योंकि वो हर महीने 1 ₹ दे रहा है।

10
9
8
7
6
5
4
3
2
1
0
55

$SI = \frac{55 \times R \times 1}{100 \times 12} = 1 \; [11 - 10]$

$\Rightarrow R = \frac{240}{11}\% = 21\frac{9}{11}\%$

II $P + \frac{PTR}{12 \times 100} = Nx + \frac{Rx}{12 \times 100}[1+2+\ldots+(n-1)]$

$x = 1,\; N = 11,\; P = 10,\; T = 11$

$10 + \frac{10 \times 11 \times R}{100} = 1 \times 11 + \frac{R \times 1}{12 \times 100}[1+2+\ldots+10]$

$\Rightarrow R = \frac{240}{11}\%$

$= 21\frac{9}{11}\%$

Compound Interest

COMPOUND INTEREST

\# Amount (A) $= P\left[1+\frac{R}{100}\right]^T$

Compound Interest (CI) $= A - P$

$$= P\left[1+\frac{R}{100}\right]^T - P = P\left\{\left(1+\frac{R}{100}\right)^T - 1\right\}$$

\# In Compound Interest principle keep on changing / compounding / updating after specific period of time according to term specified like yearly / half yearly / Quarterly etc

\# In simple words we can say that compound interest is Interest on Interest or Principle keep on compounding (updating)

\# First year simple Interest and compound Interest are equal.

$$\left[SI_{\text{first yr}} = CI_{\text{first year}}\right]$$

After one year $CI > SI$

\# If Rate of Interest are Different

$$A = P\left(1+\frac{R_1}{100}\right)\left(1+\frac{R_2}{100}\right)\left(1+\frac{R_3}{100}\right)\cdots\cdots$$

# Interest compounded	New Rate	New Time	Amount
Half yearly	$R/2$	$2T$	$P\left[1+\frac{R}{2\times100}\right]^{2T}$
Quarterly	$R/4$	$4T$	$P\left[1+\frac{R}{4\times100}\right]^{4T}$
monthly	$R/12$	$12T$	$P\left[1+\frac{R}{12\times100}\right]^{12T}$

when Interest is compounded anually but time is in fraction. e.g. $T = 2\frac{2}{5}$ yrs, then

$$\text{Amount }(A) = P\left[1+\frac{R}{100}\right]^{2}\left[1+\frac{\frac{2}{5}R}{100}\right]$$

Difference b/w simple and Compound Interest

e.g $P = 1000$, $R = 10\%$, $T = 3$ yrs

	SIMPLE INTEREST		COMPOUND INTEREST	
	P	I	P	I
1st yr.	1000	100	1000	100 → same
2ND yr.	1000	100	1100	110
3RD yr	1000	100	1210	121
		300		331

So we can see that In Simple Interest P as well as I is same for all years but in case of CI both P & I keep on updating coz in CI Interest also become principle for Next year

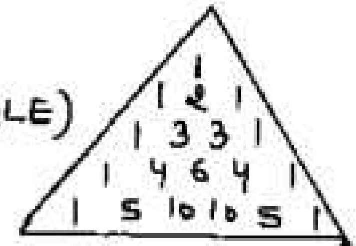
Different Techniques For Calculating CI

If $P = 20000$, $R = 10\%$ Per Annum, $T = 3$ yrs find the compound interest $CI = ?$

Technique 1 (FORMULA METHOD)

$$CI = P\left[\left(1+\frac{R}{100}\right)^3 - 1\right] = 20000\left[\left(1+\frac{10}{100}\right)^3 - 1\right] = 6620$$

Technique 2 (PASCAL'S TRIANGLE)



PASCAL Δ FOR A

CI FOR 2 yrs $= 2A + B$

CI FOR 3 yrs $= 3A + 3B + C$

CI FOR 4 yrs $= 4A + 6B + 4C + D$

CI FOR 5 yrs $= 5A + 10B + 10C + 5D + E$

where $A = R\%$ of Principle [$R\%$ is Rate]

$B = R\%$ of A

$C = R\%$ of B, $D = R\%$ of C, $E = R\%$ of D

FOR Amount $A_2 = P + 2A + B$

$A_3 = P + 3A + 3B + C$

$A_4 = P + 4A + 6B + 4C + D$

Here $CI_3 = 3[\underbrace{10\% \text{ of } 20000}_{A = 2000}] + 3[\underbrace{10\% \text{ of } 2000}_{B = 200}] + [\underbrace{10\% \text{ of } 200}_{C}]$

$= 3 \times 2000 + 3 \times 200 + 20 = 6620$ ANS.

EXAMPLE SOLVED BY PASCAL TRIANGLE

\# $P = 5000$, $R = 5\%$, $T = 3$ yrs $CI = ?$

$CI_3 = 3A + 3B + C$

$A = 5\%$ of $5000 = 250$

$B = 5\%$ of $A = 5\%$ of $250 = 12.5$

$C = 5\%$ of $B = 5\%$ of $12.5 = .625$

$\Rightarrow CI_3 = 3 \times 250 + 3 \times 12.5 + .625 = 788.125$ ANS.

\# $P = 25000$, $R = 12\%$, $T = 3$ yrs, $CI = ?$

$CI = 3(A) + 3(B) + 1(C)$

$A = 12\%$ of $25000 = 3000$

$B = 12\%$ of $A = 12\%$ of $3000 = 360$

$C = 12\%$ of $B = 12\%$ of $360 = 43.2$

$\Rightarrow CI = 3 \times 3000 + 3 \times 360 + 43.2 = 10123.2$

\# $P = 25000$, $R = 10\%$, $T = 4$ yrs, $CI = ?$

$CI_4 = 4(A) + 6(B) + 4(C) + 1(D)$

$A = 10\%$ of $25000 = 2500$

$B = 10\%$ of $A = 10\%$ of $2500 = 250$

$C = 10\%$ of $B = 10\%$ of $250 = 25$

$D = 10\%$ of $C = 10\%$ of $25 = 2.5$

$CI_4 = 4 \times 2500 + 6 \times 250 + 4 \times 25 + 2.5 = 11602.5$ ANS.

Technique 3

(i) For 2 yrs

I $P \times \frac{R}{100}$

II $P \times \frac{R}{100}$ $\quad P \times \frac{R}{100} \times \frac{R}{100}$

I $P \times \frac{R}{100}$

II $P \times \frac{R}{100}$ $\quad P \times \frac{R}{100} \times \frac{R}{100}$

→ Add up to get Simple Interest

→ Add up to get compound Interest

→ Difference b/w Simple Interest & Compound Int.

(ii) For 3 yrs

Adds Up to SI

I $P \times \frac{R}{100}$

II $P \times \frac{R}{100}$ $\quad P \times \frac{R}{100} \times \frac{R}{100}$

III $P \times \frac{R}{100}$ $\quad P \times \frac{R}{100} \times \frac{R}{100}$ $\quad P \times \frac{R}{100} \times \frac{R}{100}$ $\quad P \times \frac{R}{100} \times \frac{R}{100} \times \frac{R}{100}$

Adds Up to Difference b/w Simple Interest & Compound Interest for 3 yrs

Adds Up To get CI

$P = 12000$, $R = 10\%$, $T = 2$ yrs, $CI = ?$

I YR $1200 \xrightarrow{10\%}$

II YR 1200 120

$CI_2 = 1200 + 1200 + 120 = 2520$ Ans.

<u>Explanation</u> :-

1st yr CI = 1st yr SI = 10% of 12000 = 1200

2nd yr CI = 10% of 12000 + 10% of 1st yr CI (1200)

= 1200 + 120 = 1320

CI_2 = CI 1st YR + CI 2ND YR = 1200 + 1320 = 2520

$P = 25000$, $R = 10\%$, $T = 3$ yrs $CI = ?$

1st yr CI 2500

2ND YR CI 2500 250

3RD YR CI 2500 250 250 25

CI for 3 year = Add All = 2500 + 2500 + 2500 + 250 + 250 + 250 + 25

= 8275 ANS.

$P = 20000$, $R = 10\%$ p.a., $T = 3$ yrs, $CI = ?$

I 2000

II 2000 200

III 2000 200 200 20

Add all to get CI

$$CI_3 = 2000 + 2000 + 2000 + 200 + 200 + 200 + 20 = 6620$$

Technique 4 (Ratio/Fraction Method)

$$R = 10\% = \frac{1 \rightarrow I}{10 \rightarrow P} \Rightarrow A = P + I = 11$$

Try to assume Principle according to denominator of fraction and no. of years.

Here Assumed Principle $= (D)^n$, where

D = Denominator

n = No of years

$D = 10 \quad \{R = \frac{1}{10}\}$

$$\text{Principle}_{\text{New/Assumed}} = (10)^3 = 1000$$

$P = 1000$, $R = 10\%$, $T = 3$ yrs

I 100

II 100 + 10

III 100 + 10 + 10 + 1

$$\Rightarrow CI_3 = 331$$

By Unitary method if $P = 1000 \Rightarrow CI = 331$

but $P = 20{,}000 \Rightarrow CI_3 = \frac{331 \times 20000}{1000} = 6620$

Technique 5 (SUCCESSIVE PERCENTAGE)

$R = 10\% = \dfrac{1 \rightarrow I}{10 \rightarrow P} \Rightarrow A = 11$

	P	A
I	10	11
II	10	11
III	10	11
	1000	1331

$\underbrace{1000 \;—\; 1331}_{331 \rightarrow CI}$

IF $P = 1000$

$\Rightarrow CI = 331$

but $P = 20000$

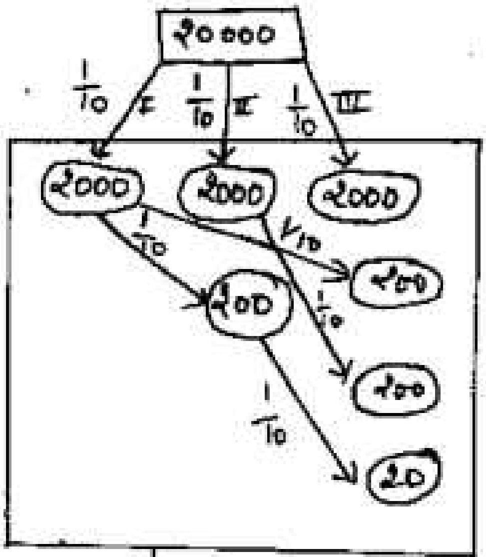
$\Rightarrow CI = \dfrac{331}{1000} \times 20000$

$= 6620$ ANS.

$\dfrac{P}{A} = \dfrac{10}{11} \times \dfrac{10}{11} \times \dfrac{10}{11}$

$P : A = 1000 : 1331$

Technique 6 (TREE METHOD)



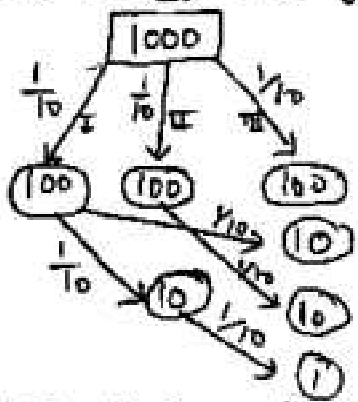
Add all these to get CI

$CI = 2000 + 2000 + 2000 + 200 + 200 + 200 + 20$

$= 6620$ ANS.

OR $R = 10\% = \dfrac{1 \rightarrow I}{10 \rightarrow P} \Rightarrow A = 11$

let $P = (10)^3 = 1000$ {T = 3 yr}



Add all present in Circle

$CI = 100 + 100 + 100 + 10 + 10 + 10 + 1$

$= 331$

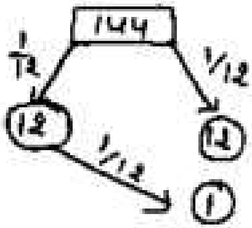
If $P = 1000 \Rightarrow CI = 331$

but $P = 20000 \Rightarrow CI = \dfrac{331}{1000} \times 20000$

$= 6620$ ANS.

\# $P = 28800$, $R = 8\frac{1}{3}\%$, $T = 2$ YR, $CI = ?$

$R = 8\frac{1}{3}\% = \frac{1 \to I}{12 \to P}$ let $P = (12)^2 = 144$ {coz $t = 2$}



OR

R		
I	12	
II	12	1

$CI_2 = 12 + 12 + 1 = 25$

Add all in circle to get $CI = 12 + 12 + 1 = 25$

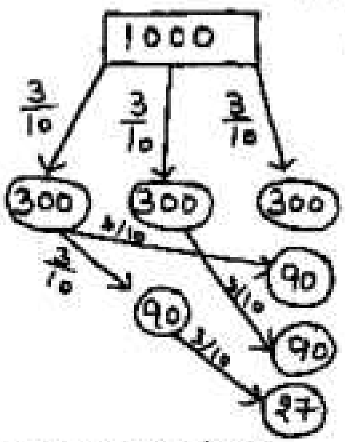
If $P = 144 \Rightarrow CI = 25$

but $P = 28800 \Rightarrow CI_2 = \frac{25}{144} \times 28800$

$= 5000$ Ans.

\# $P = 25000$, $R = 30\%$, $T = 3$ Yrs, $CI = ?$

$R = 30\% = \frac{3 \to I}{10 \to P} \Rightarrow$ let $P = (10)^3 = 1000$ {$t = 3$}



add all in circle

$CI = 1197$

OR

R				
I	300			
II	300	90		
III	300	90	90	27

$CI = 1197$

OR

R $R = \frac{3 \to I}{10 \to P} \Rightarrow A = 13$

	P	A
I	10	13
II	10	13
III	10	13
	(P) 1000	2197 (A)

$1197 \to CI$

Now If $P = 1000 \Rightarrow CI_3 = 1197$

but $P = 25000 \Rightarrow CI_3 = \frac{1197}{1000} \times 25000$

$= 29925$ Ans.

$P = 4000$, $R = 5\%$, $T = 3$ yrs, $CI = ?$

I 200

II 200 10

III 200 10 10 .5

$CI = 200 + 200 + 200 + 10 + 10 + 10 + .5$

$= 630.5$ Ans

OR $R = 5\% = \frac{1}{20} = \frac{21 \rightarrow A}{20 \rightarrow P}$

	P	A
I	20	21
II	20	21
III	20	21
(P)	8000 —	9261 (A)

$1261 \rightarrow CI$

If $P = 8000 \Rightarrow CI = 1261$

but $P = 4000 \Rightarrow CI = \frac{1261}{2} = 630.5$

$P = 3600$, $R = 16\frac{2}{3}\%$, $T = 3$ yrs $CI = ?$

$R = 16\frac{2}{3}\% = \frac{1 \rightarrow I}{6 \rightarrow P} \Rightarrow A = 7$ OR $\frac{A}{P} = \frac{7}{6}$

	P	A
I	6	7
II	6	7
III	6	7
(P)	216	343 (A)

$127 \rightarrow CI$

IF $P = 216 \Rightarrow CI = 127$

but $P = 3600 \Rightarrow CI = \frac{127 \times 3600}{216}$

$= 2116.67$ Ans.

$A = 4096$, $R = 14\frac{2}{7}\%$, $T = 2$ Yrs, $P = ?$

$R = 14\frac{2}{7}\% = \frac{1 \rightarrow I}{7 \rightarrow P} \Rightarrow A = 8$ OR $\frac{A}{P} = \frac{8}{7}$

	P	A
I	7	8
II	7	8
(P)	49	64 (A)

15

If $A = 64 \Rightarrow P = 49$

but $A = 4096 \Rightarrow P = \frac{49}{64} \times 4096$

$= 3136$ Ans

\# $P = 108000$, $R = 16\frac{2}{3}\%$, $T = 2$ yrs, $CI = ?$

$R = 16\frac{2}{3}\% = \frac{1 \rightarrow I}{6 \rightarrow P}$ let $P = (6)^2$ $\{(D)^n$, Here $D = 6\}$

$CI_1 = \frac{1}{6}$ of $36 = 6$ $= 36$ $n = 2 \downarrow$ No of years

I $6 \xrightarrow{1/6}$

II 6 1 $\Rightarrow CI_2 = 6+6+1 = 13$

if $P = 36 \Rightarrow CI_2 = 13$

but $P = 10800 \Rightarrow CI_2 = \frac{13}{36} \times 10800 = 3900$ ANS.

\# $P = 1024000$, $R = 12.5\%$, $T = 3$ yrs, $CI = ?$

$R = 12.5\% = \frac{1 \rightarrow I}{8 \rightarrow P}$ let $P = (8)^3 = 512$ $\{\because T = 3\text{ yr}\}$

$CI_1 = \frac{1}{8} \times 512 = 64$

I $64 \xrightarrow{1/8}$

II 64 $8 \xrightarrow{1/8}$

III 64 8 8 1

$CI_3 = 64+64+64+8+8+8+1$
$= 217$

$\Rightarrow$ If $P = 512 \Rightarrow CI = 217$

but $P = 1024000 \Rightarrow CI = \frac{217}{512} \times 1024000$

$= 434000$ ANS.

Explanation $\rightarrow$ Always assume $P = (D)^n$
where D = denominator of fraction, n = no of yrs

e.g If $R = 30\% = \frac{3}{10}$, $T = 3$ yrs
Assume $P = (10)^3 = 1000$

Technique 7 (Linear METHOD)

$$\underset{P}{20000} \xrightarrow[2000]{10\%\uparrow} 22000 \xrightarrow[2200]{10\%\uparrow} 24200 \xrightarrow[2420]{10\%} \underset{A_3}{26620}$$

$CI_3 = A_3 - P = 26620 - 20000 = 6620$ ANS

Technique 8 (Net Percentage Change)

$$a + b + \frac{ab}{100}$$

Here $a = b = 10\% \Rightarrow 10 + 10 + \frac{10 \times 10}{100} = 21\%$

again $a = 21,\ b = 10 \Rightarrow 21 + 10 + \frac{21 \times 10}{100} = 33.1\%$

33.1% is equivalent Rate of 3yrs

$\Rightarrow CI_3 = 33.1\%$ of $20000 = 6620$ ANS.

$P = 7500$ $R = 4\%$ $T = 2$ YRS $CI = ?$, $A = ?$

Solution:- I (Ratio)

$R\% = 4\% = \frac{1}{25} \frac{- I}{- P} \Rightarrow A = 26$

	P		A
I	25	$\rightarrow$	26
II	25	$\rightarrow$	26
	625	$\rightarrow$	676
	$\lvert \times 12$		$\lvert \times 12$
	7500		8112

$CI = A - P = 612$ ANS.

II (TRICKY)

$R = 4\% = \frac{1}{25}$

let $P = (25)^2 = (D)^2 = 625$

I 25

II 25 + 1

$\Rightarrow$ If $P = 625 \Rightarrow A = 676$

$\lvert \times 12 \Rightarrow \lvert \times 12$

7500 8112

$\Rightarrow CI = 8112 - 7500 = 612$ ANS.

$P = 20000$, $R = 5\%$, $T = 3$ YRS, $A = ?$

Solution:- I (Ratio)

$R\% = 5\% = \frac{1}{20} \frac{- I}{- P} \Rightarrow A = 21$

	P		A
I	20	$\rightarrow$	21
II	20	$\rightarrow$	21
III	20	$\rightarrow$	21
	8000	$\rightarrow$	9261
	$\lvert \times 2.5$		$\lvert \times 2.5$
	20000		23152.50

II (TRICKY)

Let $P = 5\% = \frac{1}{(20)^3} = 6000$

I 300

II 300 + 15

III 300 + 15 + 15 + 3

If $P \rightarrow 6000 \Rightarrow A = 6948$

If $P \rightarrow 20000 \Rightarrow A = \frac{6948}{6000} \times 20000$

$= 23152.50$ ANS.

$P = 1920$ $R = 12\frac{1}{2}\%$ $T = 2$ YRS $CI = ?$

let $P = \frac{N}{D} = (D)^2$ $R\% = 12.5\% = \frac{1}{8}$

$\Rightarrow$ let $P = (8)^2 = 64$

I 8

II 8 + 1

$\Rightarrow$ If $CI = 17 \Rightarrow P = 64$

$\lvert \times 30$ $\lvert \times 30$

510 ANS. 1920

$P = 20000$, $R = 10\%$, $T = 2\frac{2}{5}$, $CI = ?$, $A = ?$

$R\% = 10\% = \frac{1}{10}$ 20000

let $P = (10)^3 = 1000$

I 100

II 100 + 10

III $\boxed{100 + 10 + 10 + 1} \times \frac{2}{5}$

$\Rightarrow$ $P = 1000 \Rightarrow A = 1258.4$

$|\times 20$ $|\times 20$

20000 25168

$P = 8000$, $R = 5\%$, $T = 1$ YR 73 days $CI = ?$

Solution :- 73 days $= \frac{1}{5}$ yrs, $P = 8000$

I 400

II $\boxed{400 + 20} \times \frac{1}{5} = 84$ $\Rightarrow$ $CI_2 = 400 + 84 = \underline{484}$

$P = 50000$, $R_1 = 7\%$, $R_2 = 8\%$, $T = 2$, $CI = ?$

$R_1 = \frac{7}{100}$, $R_2 = \frac{8}{100}$ $\Rightarrow$ let $P = 2500$

I 175

II 200 + 14

$\Rightarrow$ IF $P = 2500 \Rightarrow CI = 389$

$|\times 20$ $|\times 20$

50,000 7780 ANS.

$R = 4\%$ p.a CI of 2nd year $= 208$, $P = ?$

let $P = R\% = 4\% = \frac{1}{25} = (25)^2 = 625$

I 25

II $\boxed{25 + 1} = 26$

If $CI_2 = 26 \Rightarrow P = 625$

$|\times 8$ $|\times 8$

208 $\underline{5000}$ Ans

मूलधन नहीं होते हैं

तो करते हैं

$R\% = 4\% = \frac{1}{25}$

मूलधन $= (25)^2 = 625$

625

I 25 25 II $\Rightarrow$ $26 \xrightarrow{\times 8} 208$

$P \rightarrow \underline{5000}$

R = 10%, 3rd year CI = 72.60, P = ?

Solution :- R = 10% = $\frac{1}{10}$ ⇒ P = $(10)^3$ = 1000

I 100

II 100 + 10

III $\boxed{100 + 10 + 10 + 1}$ = 121

⇒ If CI_3 = 121 ⇒ P = 1000

$|\times\frac{6}{10}$ $\quad$ $|\times\frac{6}{10}$

72.60 $\quad$ 600 Ans.

R = 5%, CI for third year = 220.50, P = ?

Solution :- R = 5% = $\frac{1}{20}$ ⇒ let P = $(20)^3$ = 8000

I 400

II 400 + 20

III $\boxed{400 + 20 + 20 + 1}$ = 441

⇒ If CI_3 = 441 ⇒ P = 8000

$|\times\frac{1}{2}$ $\quad$ $|\times\frac{1}{2}$

220.50 $\quad$ $\boxed{4000}$ Ans

R = 10%, CI for fourth year = 665.50, P = ?

Solution :- R% = $\frac{1}{10}$ ⇒ P = $(10)^4$ = 10000

I 1000

II 1000 + 100

III 1000 + 100 + 100 + 10

IV $\boxed{1000 + 100 + 100 + 10 + 100 + 10 + 10 + 1}$ = 1331

⇒ If CI_4 = 1331 ⇒ P = 10000

$|\times\frac{1}{2}$ $\quad$ $|\times\frac{1}{2}$

665.50 $\quad$ $\boxed{5000}$

CI for 2nd year = 440, R_1 = 10%, R_2 = 20%, P = ?

Solution :- let P = 2500 [LCM of = $\frac{1}{10}$ & $\frac{1}{5}$]

I 250

II $\boxed{500 + 50}$ = 550

⇒ If CI_2 = 550 ⇒ P = 2500

$|\times\frac{4}{5}$ $\quad$ $|\times\frac{4}{5}$

440 $\quad$ $\boxed{2000}$

CI FOR 3rd year = 330, $R_1 = 10\%$, $R_2 = 20\%$, $R_3 = 5\%$
$P = ?$

Solution:– Let $P = 1000$

I = 100

II 100 + 20

III $\boxed{50 + 5 + 10 + 1} = 66$

$\Rightarrow$ If $CI_3 = 66 \Rightarrow P = 1000$

$\downarrow \times 5$ $\quad$ $\downarrow \times 5$

330 $\quad$ $\boxed{5000}$ ANS.

किसी राशी पर 2 साल का SI 480 ₹ है और CI 492 ₹ है तो मूलधन और ब्याज की दर ज्ञात करें?

Solution:– SI FOR 2 year = 480

$\Rightarrow$ SI FOR 1 year = 240

SI FOR 1st year = CI FOR 1st year $= \frac{480}{2} = 240$

CI FOR IInd year = 492 − 240 = 252

Diff of CI = 252 − 240 = 12

$\therefore \frac{240 \times R \times 1}{100} = 12 \Rightarrow R = 5\%$

$\Rightarrow \frac{P \times 5 \times 1}{100} = 240$

$\boxed{P = 4800}$

किसी राशी पर 3 साल का साधारण ब्याज 1200 ₹ है और 2 साल का CI 820. Find the Rate.

Solution:– SI FOR 3 year = 1200

$\Rightarrow$ CI FOR 1st year = SI FOR 1st year $= \frac{1200}{3} = 400$

I 400

II $\boxed{400 + 20}$ $\Rightarrow R = \frac{20}{400} \times 100 = 5\%$ ANS.

$\Downarrow$

$CI_2 = 820 - 400 = 420$

\# एक व्यक्ति 390300 रु की राशि को अपने दो पुत्रों में इस प्रकार विभाजित करता है कि जब वे दोनों 18 वर्ष के हों तो उनको बराबर धनराशि प्राप्त हो जबकि उनकी आयु क्रमशः 13 वर्ष और 15 वर्ष है। यदि ब्याज की दर 4% वार्षिक चक्रवृद्धि हो तो प्रत्येक बेटे का हिस्सा बताइए

Solution :- I (BASIC)

$P_1\left[1+\frac{4}{100}\right]^5 = P_2\left[1+\frac{4}{100}\right]^3$

$P_1\left[\frac{26}{25}\right]^2 = P_2$

$\Rightarrow \frac{P_1}{P_2} = \frac{625}{676} \Rightarrow P_1 = \frac{390300}{1301} \times 625$

$= 187500$

$P_2 = \frac{390300}{1301} \times 676 = 202800.$

II (TRICKY)

Rate = 4% = $\frac{1}{25}$

1st Son	2nd Son	
$(26)^2$:	$(25)^2$	
676	625	= 1301
×300	×300	×300
202800	187500	390300

\# P = ?, R = 12.5%, T = 3 YRS

CI = 4340 (TOTAL)

Solution :- R = 12.5% = $\frac{1}{8}$

Let P = $(8)^3 = (8)^3 = 512$

I 64

II 64 8

III 64 8 8 1

CI → 64+64+64+8+8+8+1

= 217 ⇒ P = 512

20 × | ×20

4340 $\boxed{10240}$

\# R = $14\frac{2}{7}$%, T = 2 year

CI FOR IInd year = 720

P = ?

Solution :- R = $14\frac{2}{7}$% = $\frac{1}{7}$

Let P = $(7)^2 = (7)^2 = 49$

I = 7

II $\boxed{7 + 1}$ = 8

If CI_2 → 8 ⇒ P = 49

| ×90 | ×90

720 4410

Difference between CI and SI

T = 2 years

$D = \frac{PR^2}{100^2}$

T = 3 years

$D = \frac{PR^2}{100^2} \times \frac{(300+R)}{100}$

where D = Diff between CI & SI = CI − SI

D = 48, T = 2 years, R = 20%, P = ?

Solution :- I

$R = 20\% = \frac{1}{5}$

Let $P = (5)^2 = 25$

I $\boxed{5}$

II $\boxed{5} + 1 \longrightarrow D$

5I

If D = 1 ⇒ P = 25

48 × | | × 48

48 1200 Ans.

II

$R = 20\% = \frac{1}{5}$

P : D

$(5)^2 : (1)^2$

25 : 1

| × 48 | × 48

1200 48

III (FORMULA)

$D = \frac{PR^2}{100}$

$48 = \frac{P \times (20)^2}{100}$

⇒ P = 1200 Ans.

P = 2500, T = 2yrs
CI − SI = ? , R = 6%

Solution :- $R\% = \frac{6}{100} = \frac{3}{50}$

Let $P = (50)^2 = (50)^2 = 2500$

I 150

II 150 + 9 ⟶ D

II $D = \frac{PR^2}{100^2} = \frac{2500 \times (6)^2}{100^2}$

= 9

III P : D

$(50)^2 : (3)^2$

2500 : 9 → D

P = 1250, T = 2 years
CI − SI = 2 , R = ?

Solution :-

(I) Let Rate of I = R

I $1250 \times \frac{R}{100}$

II $1250 \times \frac{R}{100} + \underbrace{\boxed{1250 \times \frac{R}{100} \times \frac{R}{100}}}_{D}$

$\frac{R^2}{8} = 2 \Rightarrow R = 4\%$ Ans.

II (FORMULA) $D = \frac{PR^2}{100^2}$

$2 = \frac{1250 \times R^2}{100^2} \Rightarrow R^2 = 16$

⇒ R = 4 %

$P = 16000$, $R = 5\%$, $T = 3$ years, $CI - SI = D = ?$

Solution: (I) Let $P = (D)^3 = 8000$

I 400

II $400 + \boxed{20}$

III $400 + \boxed{20 + 20 + 1}$ — D

If $P = 8000 \Rightarrow D = 61$

$\times 2$ $\times 2$

16000 122

II (TRICKY)

$5\% = \frac{1}{20}$

P D

$D = \frac{PR^2}{100^2} \times \frac{300 + R}{100}$

$= \frac{16000 \times 5^2}{100^2} \times \frac{305}{100} = 122$ ANS

$P = 8000$, $T = 3$ years

$D = CI - SI = 61$, $R = ?$

Solution:- (TRICKY)

$\boxed{\frac{D}{P} = \frac{3R+1}{R^3}}$ $\quad \frac{61}{8000} = \frac{3R+1}{R^3}$

equating $R^3 = 8000 \Rightarrow R = 20$

rate $= \frac{1}{R} \times 100 = \frac{1}{20} \times 100 = 5\%$

$P = 40000$, $R = 5\%$, $T = 4$ YR

$D = CI - SI = ?$

Solution:- Let $P = (D)^4 = 160000$

I 8000

II $8000 + 400$

III $8000 + 400 + 400 + 20$

IV $8000 + 400 + 400 + 20 + 400 + 20 + 20 + 1$

If $P = 160000 \Rightarrow D = 2481$

$\times \frac{1}{4}$ $\times \frac{1}{4}$

40000 620.25 ANS

$P = 2000$, $T = 2$ years

$R_1 = 5\%$, $R_2 = 4\%$

$D = CI - SI = ?$

Solution:- $P = 2000$

I 100

II $80 + 4 \rightarrow D$

$P = 5000$, $T = 2$ years

$D = CI - SI = 25$, $R_1 = 10\%$

$R_2 = ?$

Solution:- 5000

I 500

II $\frac{5000 \times R_2}{100} + \boxed{\frac{500 \times R_2}{100}}$ — D

$\frac{500 \times R_2}{100} = 25 \Rightarrow R_2 = 5\%$

Type 2 (when Rate is Not given)

TRICK Explanation

$T = 2$ yrs

$$A = P\left[1+\frac{R}{100}\right]^2$$

$$\frac{A}{P} = \left[1+\frac{R}{100}\right]^2$$

$$\sqrt{\frac{A}{P}} = \left[1+\frac{R}{100}\right]$$

$T = 3$ yrs

$$A = P\left[1+\frac{R}{100}\right]^3$$

$$\frac{A}{P} = \left[1+\frac{R}{100}\right]^3$$

$$\sqrt[3]{\frac{A}{P}} = \left[1+\frac{R}{100}\right]$$

1 year change (SI = CI)

$\left[1+\frac{R}{100}\right]$ →

- $T = 2$ yrs $\Rightarrow \sqrt{\frac{A}{P}}$ (वर्गमूल कर देंगे)
- $T = 3$ yrs $\Rightarrow \sqrt[3]{\frac{A}{P}}$ (घनमूल कर देंगे)
- $T = 4$ yrs $\Rightarrow \sqrt[4]{\frac{A}{P}}$
- $T = n$ yrs $\Rightarrow \sqrt[n]{\frac{A}{P}}$

e.g. if $1+\frac{R}{100} = \frac{4}{3} \Rightarrow \frac{R}{100} = \frac{1}{3}$ OR $R = 33.33\%$

OR $1+\frac{R}{100} = \frac{4 \rightarrow A_1}{3 \rightarrow P_1}$ $\quad I = 1 \quad R\% = \frac{1}{3} \times 100$

$= 33.33\%$

A sum of money becomes 9 times in 2 years at a certain rate of CI. Find the rate?

Solution :- I (BASIC)

Let $P = 1 \Rightarrow A = 9$

$$9 = 1\left[1+\frac{R}{100}\right]^2$$

OR $1+\frac{R}{100} = 3$

$\frac{R}{100} = 2 \Rightarrow R = 200\%$ **ANS.**

II (TRICKY)

T = 2 yrs take square Root

$$\left[1+\frac{R}{100}\right] = \sqrt{\frac{9}{1}} = \frac{3}{1}$$

$\Rightarrow \frac{R}{100} = 2 \Rightarrow R = 200\%$

OR $\sqrt{9} : \sqrt{1} = \underset{A}{3} : \underset{P}{1}$

$R\% = \frac{2}{1} \times 100 = 200\%$

कोई राशी CI की दर से 2 साल में 9/4 गुणी हो जाती है। ब्याज की दर ज्ञात करो?

Solution T = 2 yrs, Let P = 1 $\Rightarrow A = \frac{9}{4}$

$$1+\frac{R}{100} = \sqrt{\frac{9}{4}} = \frac{3}{2}$$

$\Rightarrow \frac{R}{100} = \frac{1}{2} \Rightarrow \boxed{R = 50\%}$

OR

A	P
9	4

T = 2 yrs take sq root

$\sqrt{9}$: $\sqrt{4}$

$\underbrace{3 \quad 2}_{I=1}$

में 1 साल का Ratio आ जाता है। अब SI को use करो.

$R\% = \frac{1}{2} \times 100 = 50\%$

₹ 8000 becomes 9261 in 3 years at certain rate of compound interest. Find the Rate?

Solution :- I

T = 3 yrs take cube Root

$$1+\frac{R}{100} = \sqrt[3]{\frac{9261}{8000}} = \frac{21}{20}$$

$\frac{R}{100} = \frac{1}{20} \Rightarrow \boxed{R = 5\%}$

में एक साल का P:A आ जाता है। अब हम SI लगा सकते हैं।

II

A		P
9261	:	8000

T = 3 yrs take cube ROOT

$\sqrt[3]{9261} : \sqrt[3]{8000}$

$\underbrace{21 : 20}_{CI = 1}$

$R\% = \frac{1}{20} \times 100 = 5\%$

A sum of money becomes 25 times in 5 yrs and 36 times in 7 years. Find the Rate of interest?

Solution :— I (BASIC)

Let P = 1₹

$\therefore 25 = \left[1+\frac{R}{100}\right]^5$ — ①

also $36 = \left[1+\frac{R}{100}\right]^7$ — ②

divide ② by ①, we get

$\left[1+\frac{R}{100}\right]^2 = \frac{36}{25} \Rightarrow 1+\frac{R}{100} = \frac{6}{5}$

$\Rightarrow R = 20\%$ ANS

II (TRICKY)

2 yrs change $\begin{cases} A_7 \text{ (7 yrs)} \rightarrow 36 \\ A_5 \text{ (5 yrs)} \rightarrow 25 \end{cases}$

$\Rightarrow$ 1 yr change $= \sqrt{\frac{36}{25}} = \frac{6}{5}\frac{A}{P}$

A : P (For 1 yr)

$6 : 5$, $\underbrace{\ }_{1}$

$\Rightarrow R\% = \frac{1}{5} \times 100$
$= 20\%$

कोई राशी CI की दर से 2 साल में 8 गुणी हो जाती है। और 5 साल में 27 गुणी हो जाती है तो ब्याज की दर ज्ञात करो?

Solution :—

3 yrs change $\leftarrow \begin{cases} A_5 \longrightarrow 27 \\ A_2 \longrightarrow 8 \end{cases}$

$\Rightarrow$ In 1 year $\rightarrow$ $\overset{A}{\sqrt[3]{27}} : \overset{P}{\sqrt[3]{8}} = \overset{A}{3} : \overset{P}{2}$, $\underbrace{\ }_{1}$

$R\% = \frac{1}{2} \times 100 = 50\%$

कोई राशी CI की दर से 4 साल में 16000 और 7 साल में 18522 हो जाती है। ब्याज की दर ज्ञात करें?

Solution :—

3 yrs change $\leftarrow \begin{cases} A_7 \text{ — } 18522 \\ A_4 \text{ — } 16000 \end{cases}$

$\Rightarrow$ 1 yr change $\overset{A}{\sqrt[3]{18522}} : \overset{P}{\sqrt[3]{16000}} = \overset{A}{21} : \overset{P}{20}$, $\underbrace{\ }_{1}$

$R\% = \frac{1}{20} \times 100 = 5\%$

If a sum of money becomes N_1 Times in T_1 yrs and N_2 times in T_2 years then

$$(N_1)^{T_2} = (N_2)^{T_1} \quad \text{OR} \quad N_2 = (N_1)^{T_2/T_1}$$

A sum of money get doubled in 4 years. In how much time it will become 8 times?

Solution:— I (FORMULA)

$N_1 = 2,\ T_1 \to 4$

$N_2 \to 8,\ T_2 = ?$

$N_2 = (N_1)^{T_2/T_1} \Rightarrow 8 = (2)^{T_2/4}$

OR $2^3 = (2)^{T_2/4} \Rightarrow T_2/4 = 3$

$\Rightarrow T_2 = 12$ yrs.

II (TRICKY)

2^1 गुनी $\longrightarrow 4$
2^2 times $\longrightarrow 8$]+4
2^3 times $\longrightarrow 12$]+4
8 times — 12 yrs

कोई राशि चक्रवृद्धि ब्याज की दर से 3 साल में 3 गुनी हो जाती है। कितने समय में यह राशि 81 गुनी हो जाएगी?

Solution:— I (FORMULA)

$N_1 \to 3 \quad T_1 - 3$

$N_2 - 81 \quad T_2 \to ?$

$N_2 = (N_1)^{T_2/T_1}$

$\Rightarrow 81 = (3)^{T_2/3} \Rightarrow 3^4 = 3^{T_2/3}$

$\Rightarrow T_2/3 = 4 \Rightarrow 12$ yrs

II (TRICKY)

3^1 times —— 3
3^2 times —— 6]+3
3^3 times —— 9]+3
$3^4 = 81$ times $\longrightarrow$ 12]+3

12 years Ans.

A Sum of money becomes 81 times in 4 years. in how much time it will become 27 times?

Solution:— I (FORMULA)

$N_1 - 81,\ T_1 - 4$

$N_2 - 27,\ T_2 - ?$

$27 = (81)^{T_2/4}$

OR $\Rightarrow 27 = (3^4)^{T_2/4}$

$3^3 = 3^{T_2} \Rightarrow T_2 = 3$ yrs.

II (TRICKY)

	A	:	P
$T=4 \to$	81	:	1
$T=1 \to$	$\sqrt[4]{81}$	:	$\sqrt[4]{1}$
	3	:	1

$\therefore 3^1$ —— 1 साल
3^2 —— 2 साल
$27 = 3^3$ —— 3 साल $\to$ Ans 3

$P = 625$ ₹ $\quad R = 4\% \quad T = 2$ yrs, $CI = ?$

$4\% = \frac{1 \text{ — } I}{25 \text{ — } P} \Rightarrow A = 26$

$$\begin{array}{c} 25 : 26 : 26 \\ 25 : 25 : 26 \\ \hline 625 : 650 : 676 \end{array}$$

मूलधन ← 625

650 → 1 साल के बाद मिश्रधन

676 → 2 साल के बाद मिश्रधन

$\boxed{25 \text{ ₹}}$ पहले साल का ब्याज

$\boxed{26 \text{ ₹}}$ दूसरे साल का ब्याज

$\boxed{1 \text{ ₹}}$ दो साल के ब्याज का अन्तर

$I = A - P$

$= 676 - 625 = 51$

$1000 = P$, $R = 10\%$, $T = 3$ yrs

CI

$10\% = \frac{1}{10}$

$$\begin{array}{cccc} 10 : & 11 : & 11 : & 11 \\ 10 : & 10 : & 11 : & 11 \\ 10 & 10 : & 10 : & 11 \\ \hline 1000 & & & 1331 \end{array}$$

331

ANS.

$P = 3000$, $R = 20\%$

$T = 2$ yrs, $A = ?$

$20\% = \frac{1}{5}$

$$\begin{array}{ccc} 5 : & 6 : & 6 \\ 5 : & 5 : & 6 \\ \hline (P)\ 25 : & 30 : & 36\ (A) \end{array}$$

$25 \xrightarrow{\times 120} 3000$

$36 \xrightarrow{\times 120} 4320$ ANS.

A sum of money becomes 4800 after 4 years and ₹ 6000 after 8 years. Find the Principal?

Solution:– I (TRICKY)

$A_8 \longrightarrow 6000$

$A_4 \longrightarrow 4800$

4 yr change $\frac{A_8}{A_4} = \frac{6000}{4800} = \frac{5}{4}$

[Every 4 yrs it will change by 5/4]

$\therefore \frac{A_4}{P} = \frac{5}{4} \Rightarrow \frac{4800}{P} = \frac{5}{4} \Rightarrow P = 3840$

OR $P \times \frac{5}{4} = 4800$ OR $P \times \frac{5}{4} \times \frac{5}{4} = 6000$

$P = 3840$ ANS.

II (SUPER TRICKY)

$P \overset{4}{\frown} A_4 \overset{4}{\frown} A_8$

$$
\begin{array}{ccccc}
 & & 4800 & : & 6000 \\
4 & : & 4 & : & 5 \\
4 & : & 5 & : & 5 \\
\hline
16 & : & 20 & : & 25 \\
\downarrow \times 240 & & \downarrow \times 240 & & \\
3840 & & 4800 & &
\end{array}
$$

$\underline{3840}$

कोई धनराशि 8 साल के बाद 1600 ₹ हो जाती है और 12 साल के बाद 2000 ₹ हो जाती है तो 16 साल बाद कितनी हो जायेगी?

Solution:– I

$A_{12} \longrightarrow 2000$

$A_8 \longrightarrow 1600$

4 yrs

4 साल में $= \frac{2000}{1600} = \frac{5}{4}$ गुनी

(हर 4 साल बाद $\frac{5}{4}$ गुनी हो जायेगी]

$A_{12} \times \frac{5}{4} = A_{16}$

$2000 \times \frac{5}{4} = 2500$ ₹ ANS

II

$A_8 \overset{4\,Yr}{\frown} A_{12} \overset{4\,Yr}{\frown} A_{16}$

$$
\begin{array}{ccccc}
1600 & : & 2000 & & \\
4 & : & 5 & : & 5 \\
4 & : & 4 & : & 5 \\
\hline
16 & : & 20 & : & 25 \\
 & & \downarrow \times 100 & & \downarrow \times 100 \\
 & & 2000 & & 2500
\end{array}
$$

2500 ANS

\# CI for 2 years and 3 years for a certain sum is 156 and 254 respectively. Find the rate of compound interest? a) $14\frac{2}{7}\%$ b) $16\frac{2}{3}\%$ c) 10% d) 9%.

Solution:- $\dfrac{CI_3 = P\left[\left(1+\frac{R}{100}\right)^3-1\right]}{CI_2 = P\left[\left(1+\frac{R}{100}\right)^2-1\right]} = \dfrac{254}{156} = \dfrac{127}{78}$ — (1)

Let $1+\frac{R}{100} = x \Rightarrow \dfrac{x^3-1}{x^2-1} = \dfrac{127}{78} \Rightarrow \dfrac{(x-1)(x^2+1+x)}{(x-1)(x+1)} = \dfrac{127}{78}$

$\Rightarrow 78x^2+78+78x = 127x+127$

$\Rightarrow 78x^2-127x+78x+78-127=0 \Rightarrow 78x^2-49x-49=0$

$78x^2-91x+42x-49=0$

or $13x[6x-7]+7[6x-7]=0 \Rightarrow x=\frac{7}{6}$ ∴ $1+\frac{R}{100} = \frac{7}{6}$

$\Rightarrow R = 16\frac{2}{3}\%$ ANS.

OR check by option in (1)

option b will satisfy.

\# Find the compound interest on Rs 15625 for 9 months at 16% p.a. compounded quarterly?

Solution:- $R = \frac{16}{4} = 4\%$ and $T = 9\times4 = 36$ months $= 3$yrs

$4\% = \frac{1}{25}$

P	A
25	26
25	26
25	26
15625	17576

CI = A − P = 1951

OR PASCAL 3 3 1

3A + 3B + C

$\dfrac{3\times15625\times4}{100} + \dfrac{3\times625\times4}{100} + 1$

$= 625\times3 + 25\times3 + 1$

$= 1951$ ANS.

INSTALLMENT (किश्त) IN COMPOUND INTEREST

$$P = \frac{x}{\left[1+\frac{R}{100}\right]^1} + \frac{x}{\left[1+\frac{R}{100}\right]^2} + \cdots\cdots + \frac{x}{\left[1+\frac{R}{100}\right]^T}$$

where P = मूलधन / Principal, x = किश्त (INSTALLMENT)

T → समय (Time) OR No. of Installments

A man borrows Rs 21000 at 10% compound interest. How much he has to pay equally at the end of year to settle his loan in two years.

Solution :- I (FORMULA)

$$21000 = \frac{x}{\left[1+\frac{10}{100}\right]} + \frac{x}{\left[1+\frac{10}{100}\right]^2}$$

OR $21000 = \frac{10x}{11} + \frac{100x}{121}$

OR $21000 = \frac{110x+100x}{121}$

OR $x = 12100$ Ans.

II (TRICKY)

$R = 10\% = \frac{1}{10} \rightarrow \frac{I}{P} \Rightarrow$ Installment $= 11$

[make each installment equal]

	P	Installment
1st year	$10_{\times 11}$	$11_{\times 11}$
2nd year	$(10)^2$	$(11)^2$

OR

110	121
100	121

$21000 = 100 \times 210$ $121 \times 100 = 12100$

A man buys a scooter on making a cash down payment of ₹ 16224 and promise to pay two more yearly installment of equivalent amount in next two years. If the Rate of Interest is 4% p.a. C.I. Find the cash value of scooter?

Solution :- I (BASIC)

$$P = \frac{16224}{\left(1+\frac{4}{100}\right)^1} + \frac{16224}{\left[1+\frac{4}{100}\right]^2}$$

$$= 16224\left[\frac{25}{26} + \frac{625}{676}\right]$$

$$= 30600$$

cash value $= 30600 + 16224$

$= 46824$ Ans.

II (TRICKY) $R = 4\% = \frac{1}{25}$

	Principal	Installment
I	$25_{\times 26}$	$26_{\times 26}$
II	+ 625	676
	1275	676
	1×24	1×24
	30600	16224

CASH VALUE $= 30600 + 16224 = 46824$

A man borrows Rs 4200 and undertakes to pay back with CI @ 10% p.a in two equal yearly installments at the end of 1st and 2nd year. What is the amount of each installment?

Solution :- $R = 10\% = \frac{1}{10}$ [but installment must be equal]

	P	Inst.
1st Yr	10×11	11×11
2nd Yr	+ 100	121
	210	121
	$\times 20$	$\times 20$
	4200	2420 Ans.

OR by FORMULA

$$4200 = \frac{x}{\left[1+\frac{10}{100}\right]} + \frac{x}{\left[1+\frac{10}{100}\right]^2}$$

$$\Rightarrow 4200 = x\left[\frac{10}{11} + \frac{100}{121}\right]$$

$$\Rightarrow 4200 = \frac{210x}{121} \Rightarrow x = 2420 \text{ Ans.}$$

A man borrows Rs 1820 and undertakes to pay back with compound interest @ 20% p.a in 3 equal yearly installment at the end of 1st, 2nd and 3rd year. What is the amount of each installment?

Solution :- $R = 20\% = \frac{1}{5 - P} \Rightarrow \text{Installment} = 5 + 1 = 6$

1st	5×36	$6 \times 36 = 216$	Installment must be equal
2nd	25×6	$36 \times 6 = 216$	
3rd	125	$216 = 216$	
	455	216	
	$\times 4$	$\times 4$	
	1820	864 Ans.	

A man borrows a certain sum of money and pays it back in two equal installments, if CI is reckoned at 5% per annum and he pays back annually Rs 882. What sum did he borrow?

Solution :- $R = 5\% = \frac{1}{20}$

	P	I	
1st	20×21	21×21	[Installment must be equal]
2nd	+ 400	441	
	820	441	
	$\times 2$	$\times 2$	
Ans	1640	882	